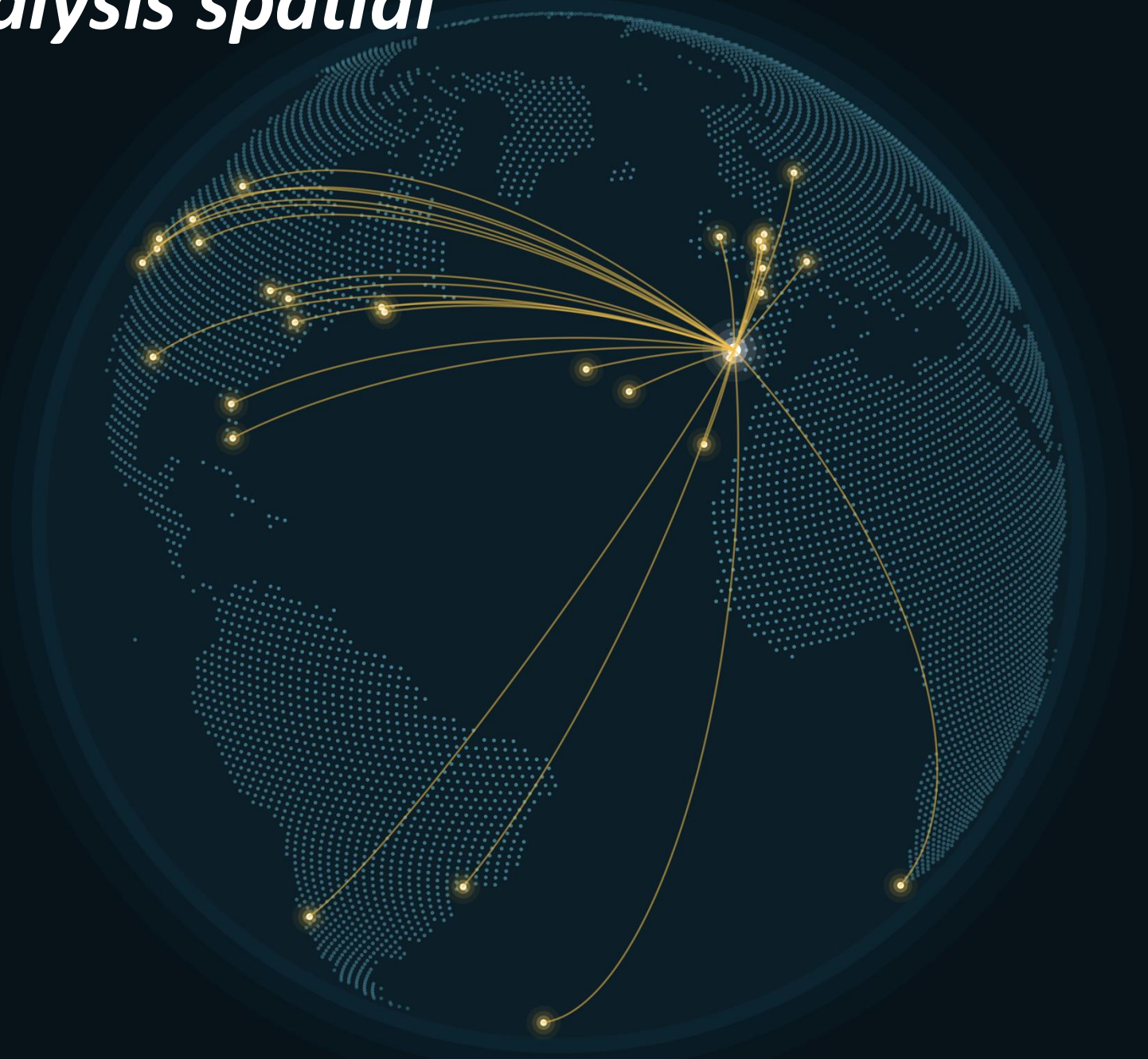
Making your meta-analysis spatial

Where does our evidence come from, and why should we care?



Jorge Pereira

(jorge.pereira@uc.pt)

1 2 9 0 CAMPUS FIGUEIRA DA FOZ UNIVERSIDADE D COIMBRA

CENTRE FOR FUNCTIONAL ECOLOGY
SCIENCE FOR PEOPLE & THE PLANET

Outline of the session



Outline of the session



1. Spatial data: the (very) basics

Why spatial thinking matters

2. Spatial data in meta-analysis

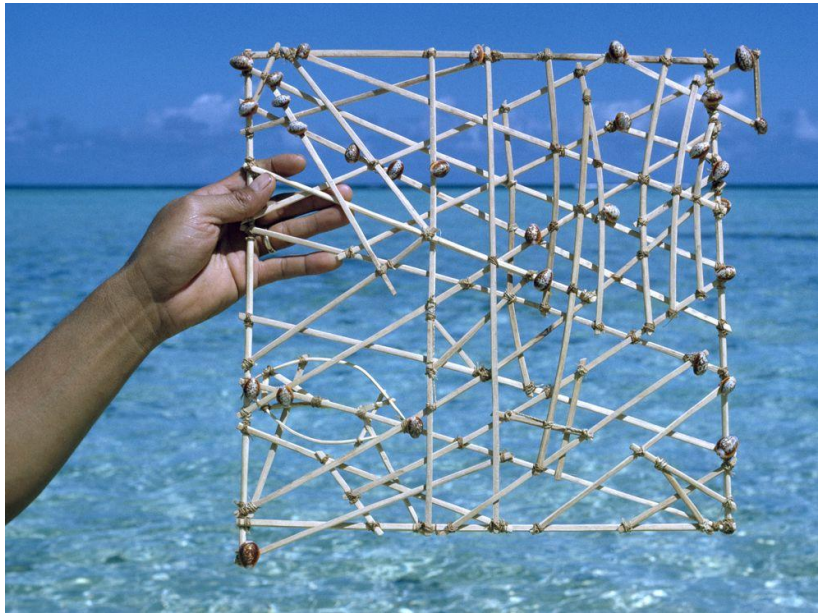
Why map a meta-analysis?
Case study: seabirds and environmental change
Check the raw data first
Where does the evidence come from?
Who does the research?
When were the colonies tracked?
What did the studies find?
Why does the effect change across space?
Which studies are not independent?

3. Practical session

Mapping Alfredo noisy birds
Mapping your own data

Humans have always made maps

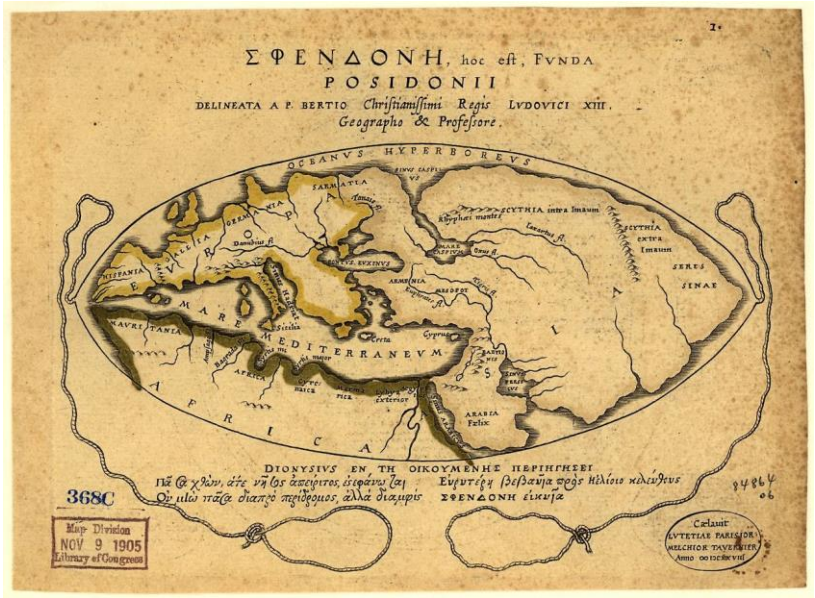
- People have created maps for thousands of years
- Maps guided exploration, trade and resource management



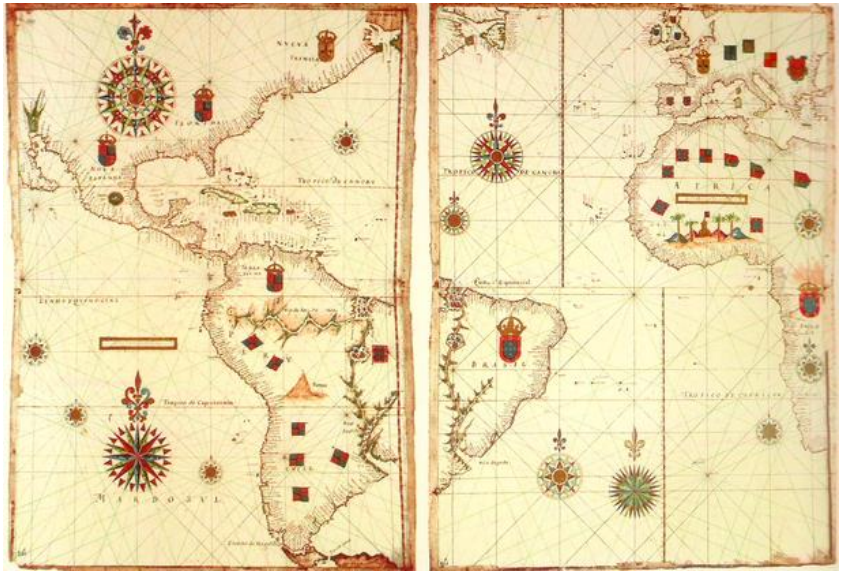
Marshall Islands stick chart (> 1000 years)



Babylonian world map (c. 600 BC)



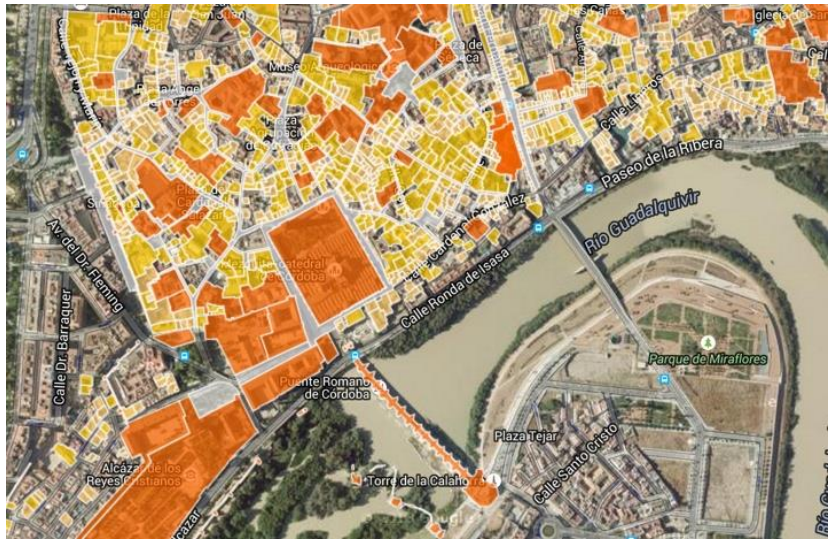
Ancient Greece (c. 100 BC)



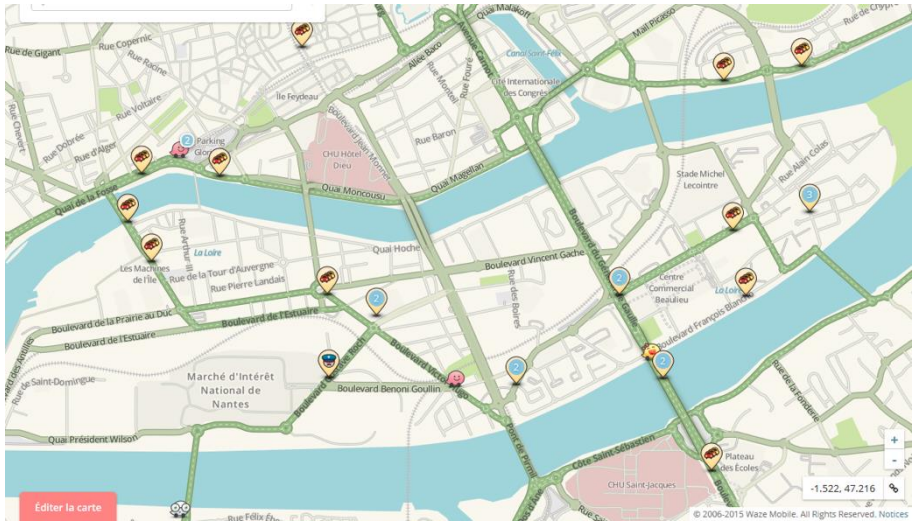
Map of the Tordesillas division (1494)

Maps are everywhere

- Today we use them daily: navigation, weather, disease spread, crop monitoring, etc
- Most data (e.g. biological and health) have a location
- Your results depend on where the data come from



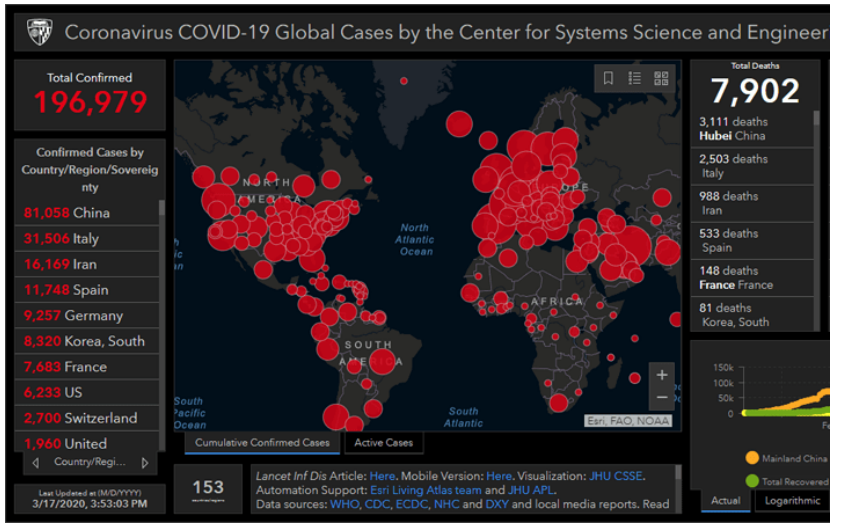
Urban mapping (Google Earth)



Real-time navigation (Waze or Google Maps)

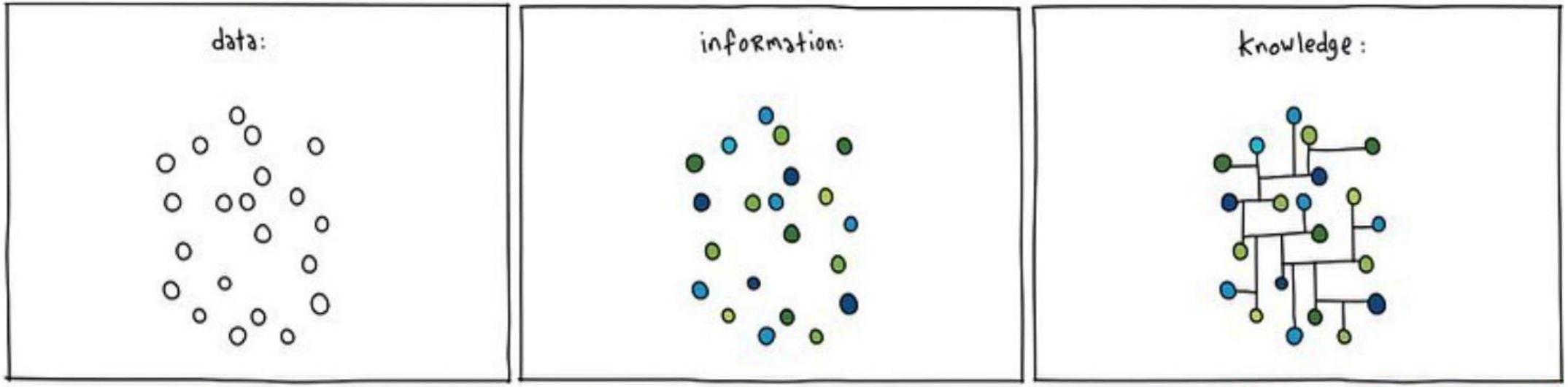


Satellite flooding map (València, 2024)

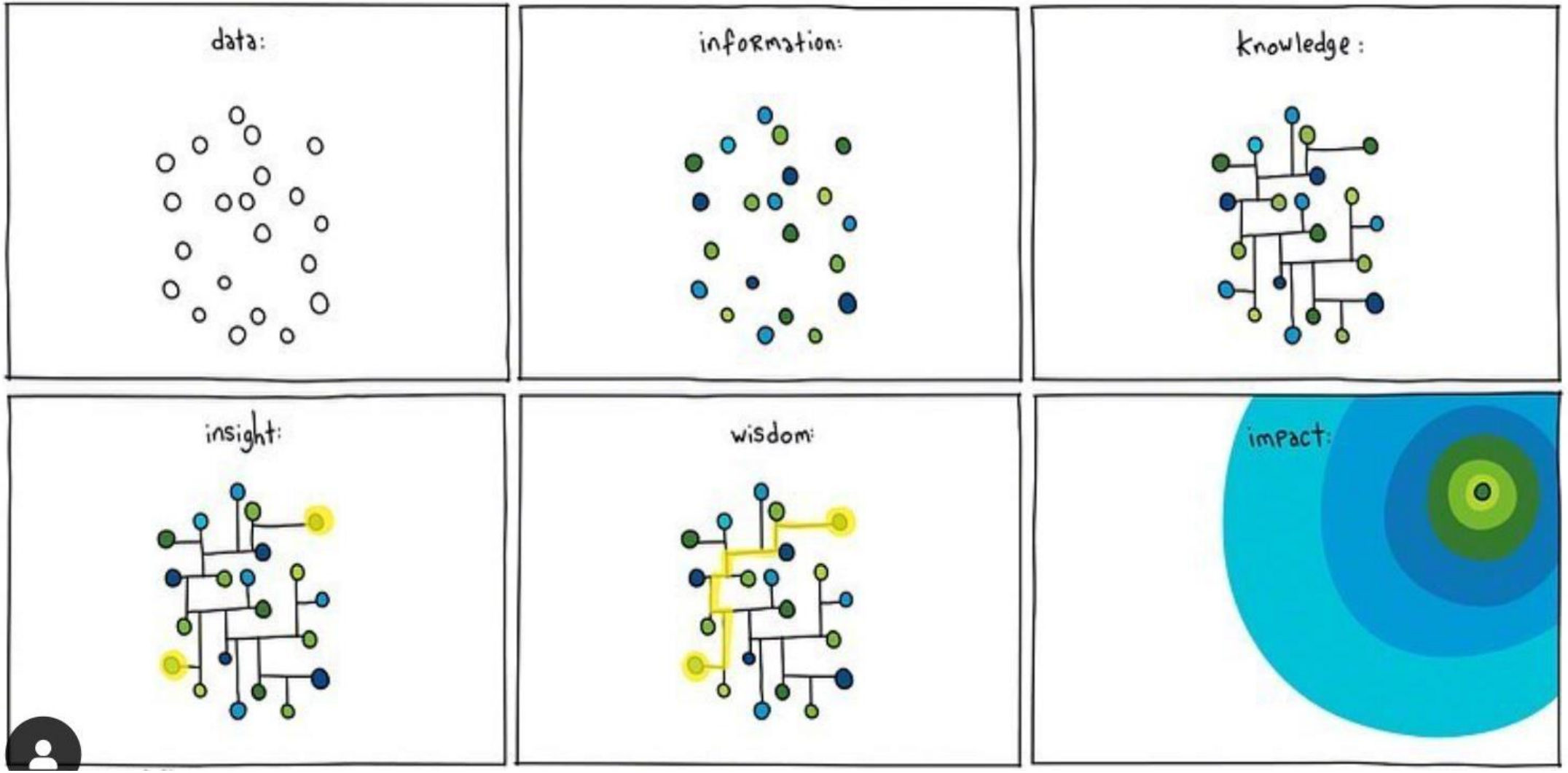


COVID-19 daily dashboard (lockdown, 2020)

Importance of thinking geographically

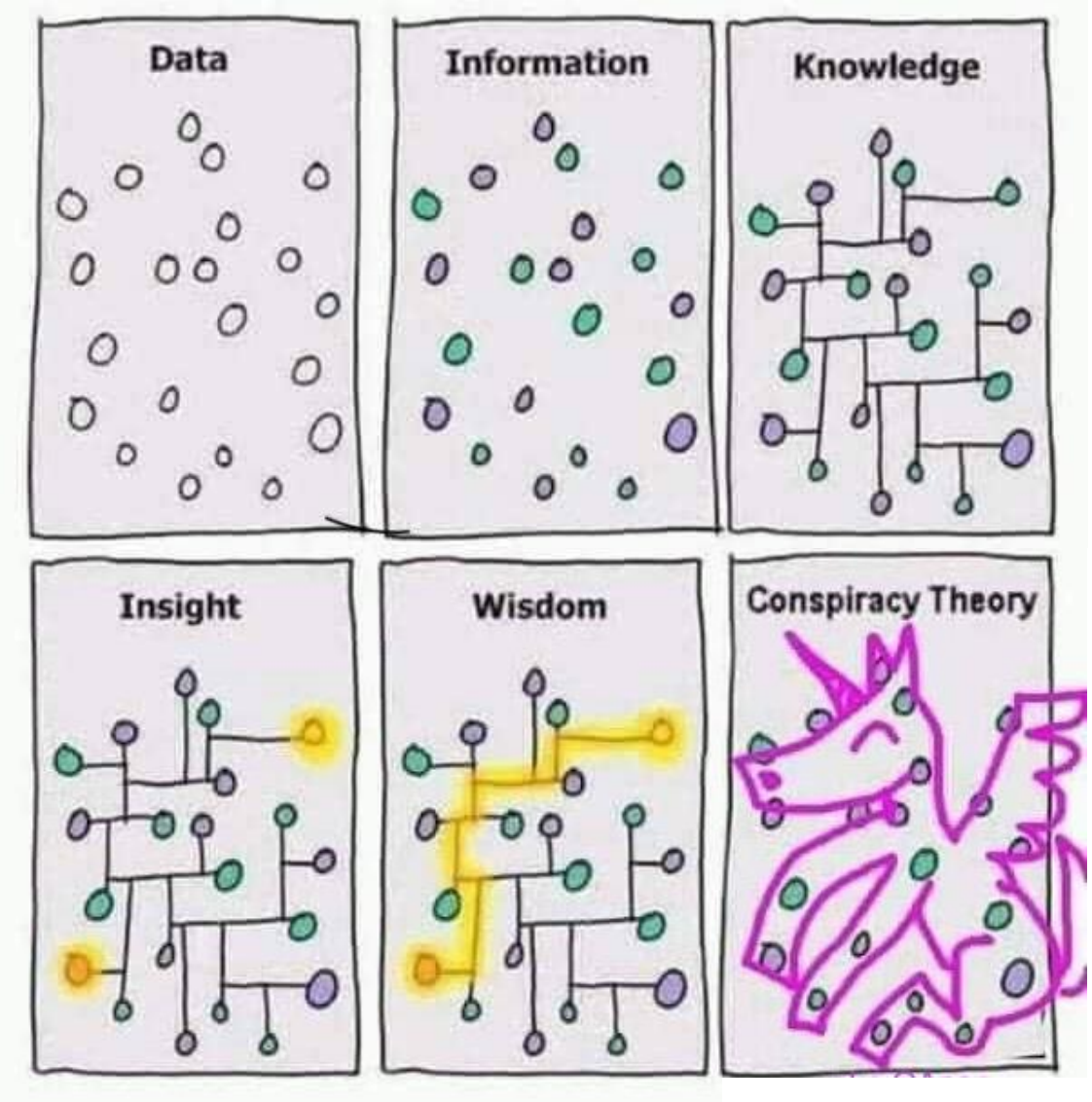


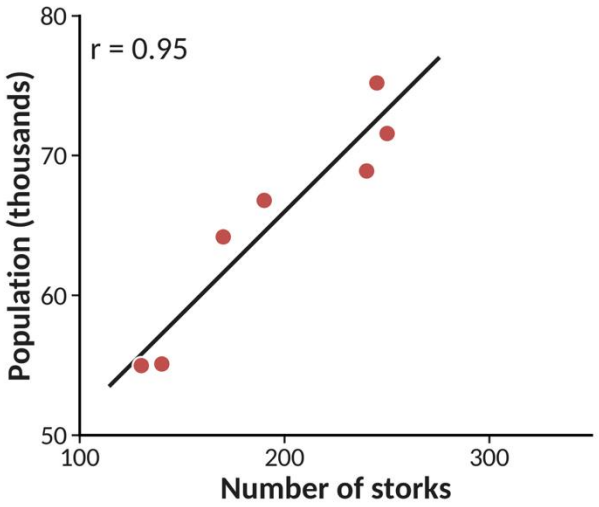
Importance of thinking geographically

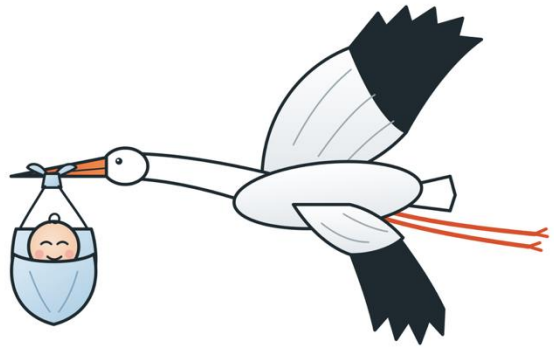


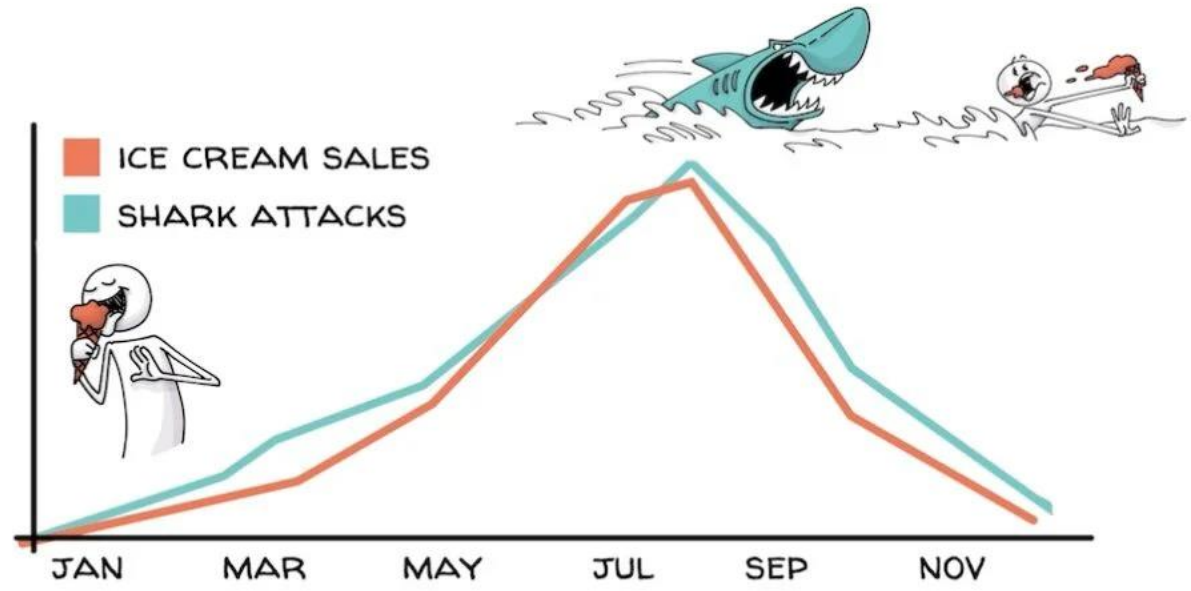
"Any fool can know. The point is to understand."

You can embrace chaos...

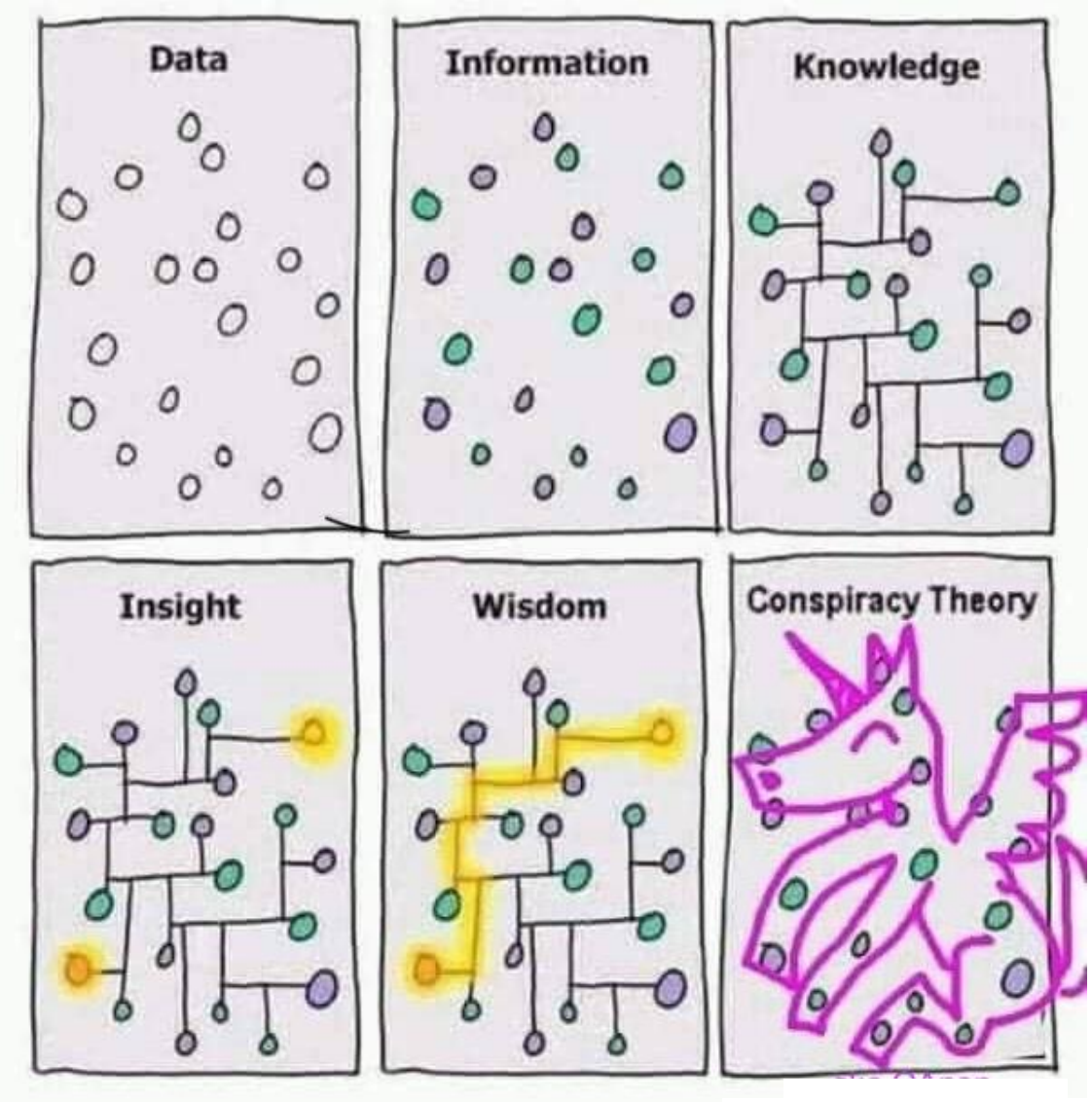






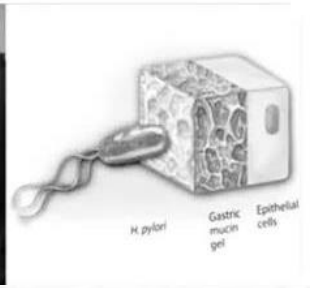


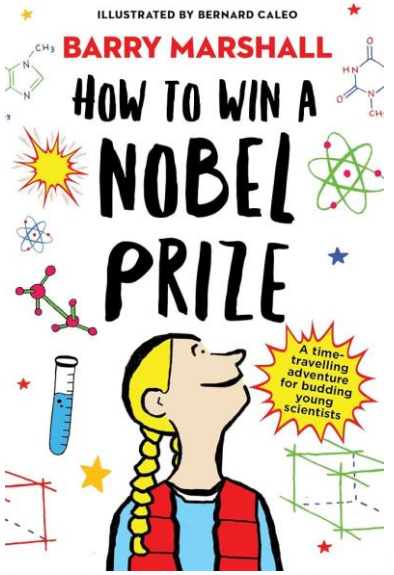
You can embrace chaos... or creativity

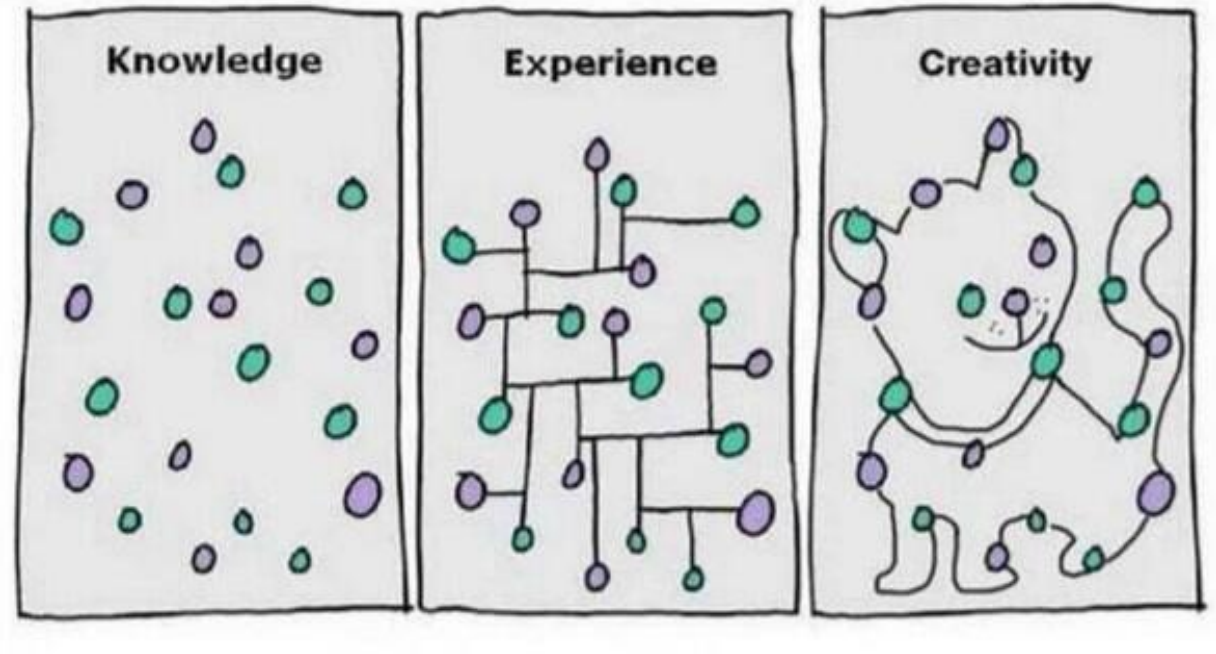


Dr. Barry J. Marshall was convinced that H. pylori bacteria causes stomach ulcers, but no one believed him. Since it was illegal to test his theory on humans, he drank the bacteria himself, developed ulcer within days, treated it with antibiotics, and won a Nobel Prize.





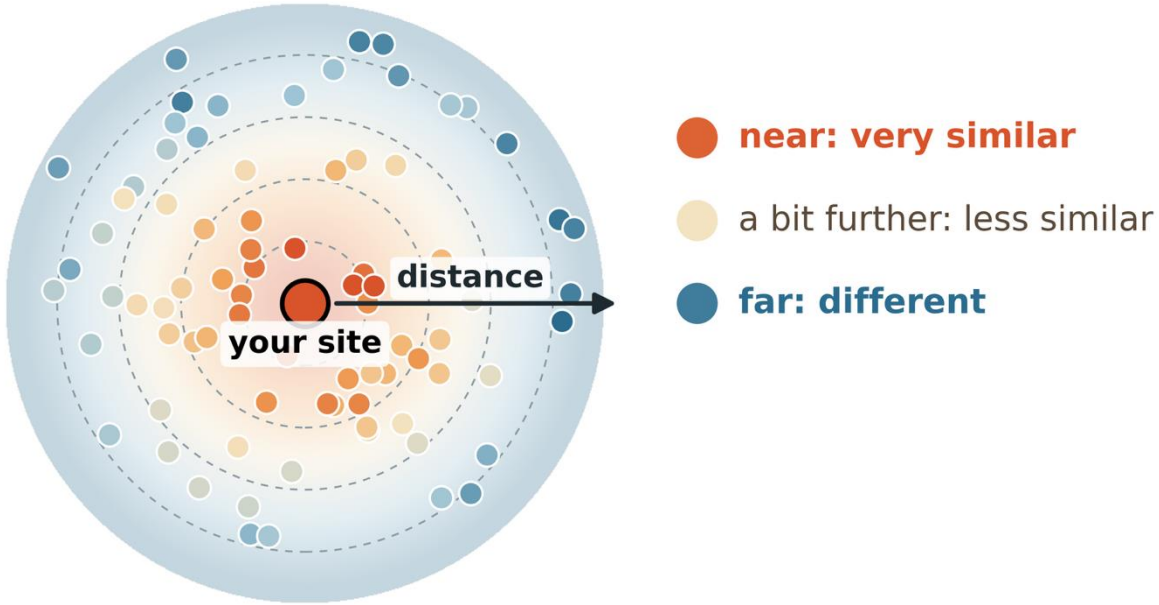




Spatial thinking is special

Tobler's First Law of Geography:

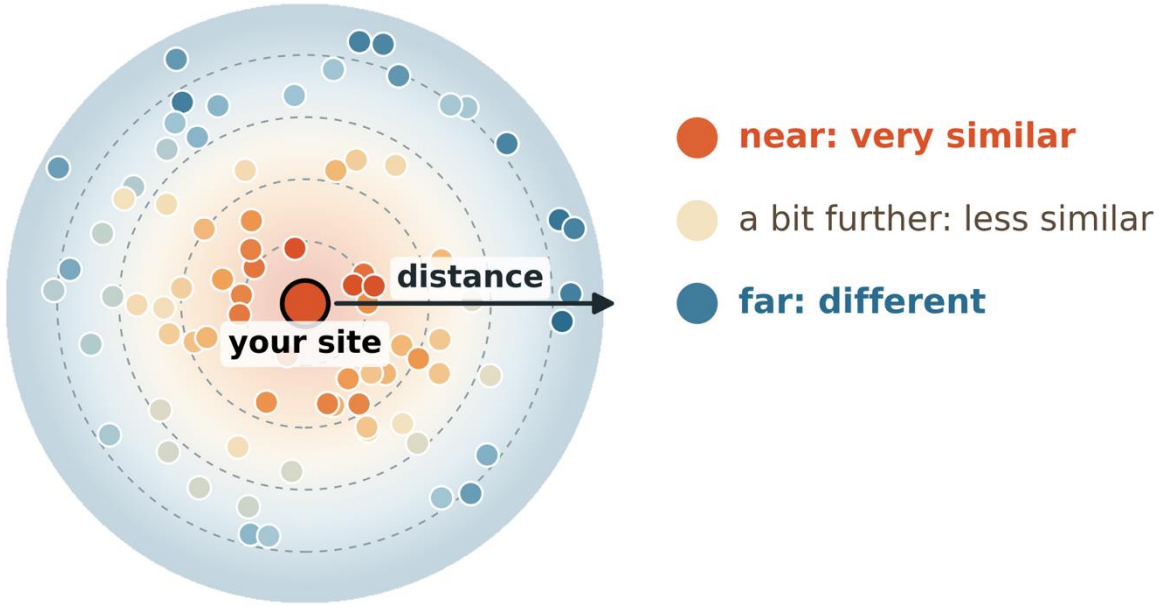
"Everything is related to everything else, but near things are more related than distant things."



Spatial thinking is special

Tobler's First Law of Geography:

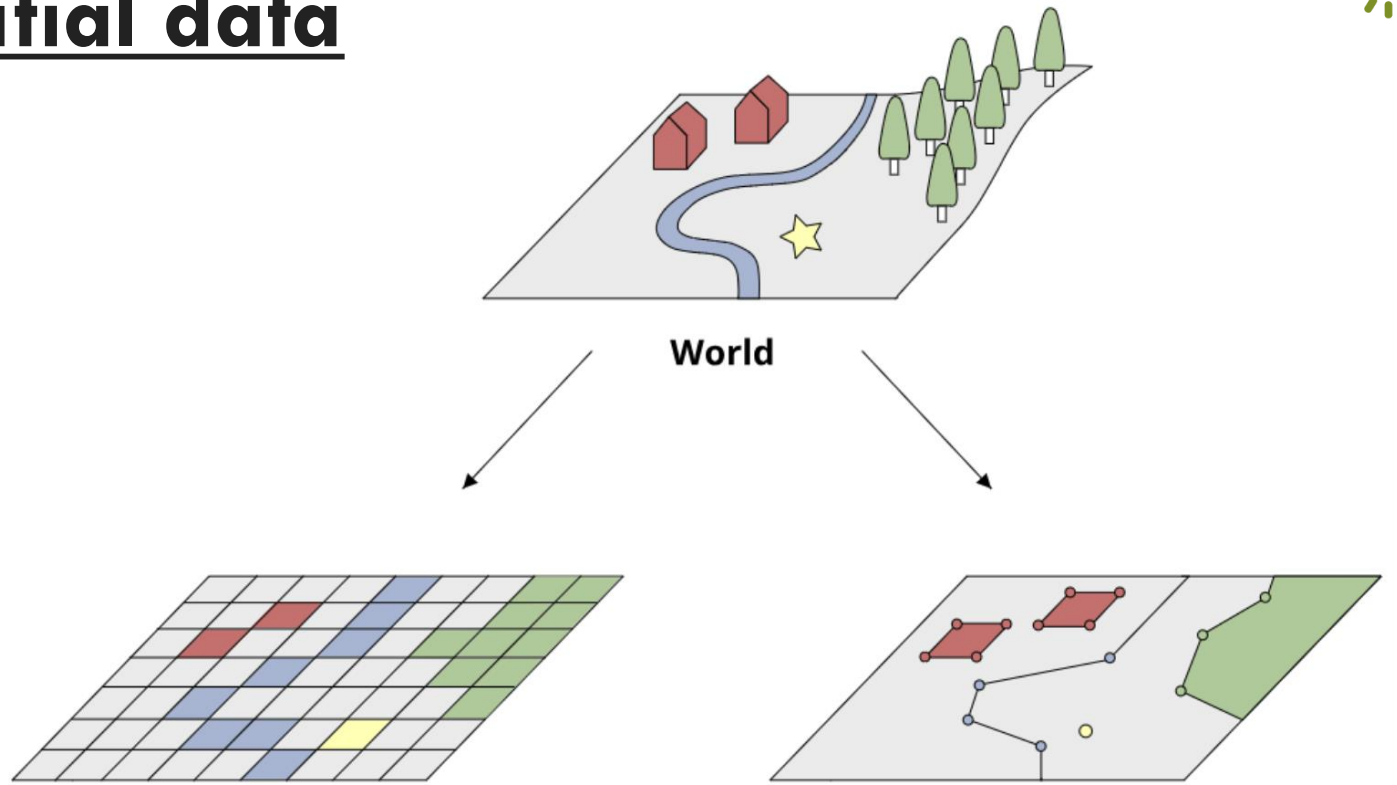
"Everything is related to everything else, but near things are more related than distant things."







Types of spatial data



Raster data | each cell stores a value

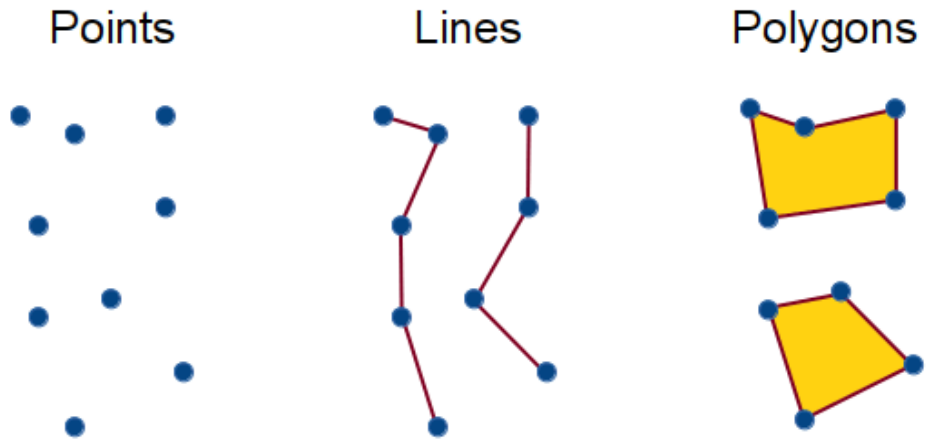
- Categories (integer)
- Continuous (decimal)
- Reflectance (satellite images)

Vector data | geometry and attributes

- Location: where the feature is (x, y)
- Geometry: points, lines or polygons
- Attributes: what do we know about it

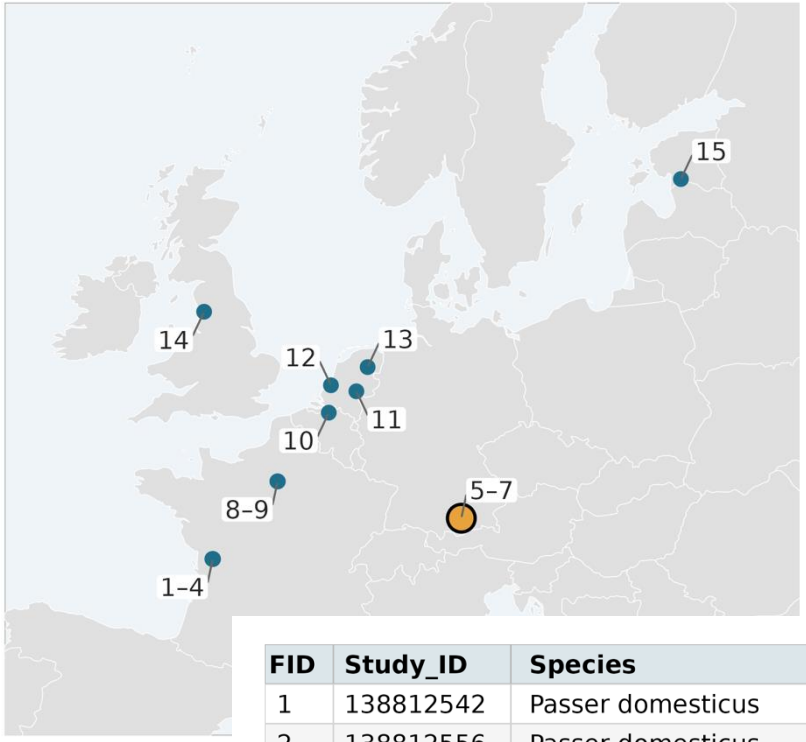
Vector data

Spatial data



In reality

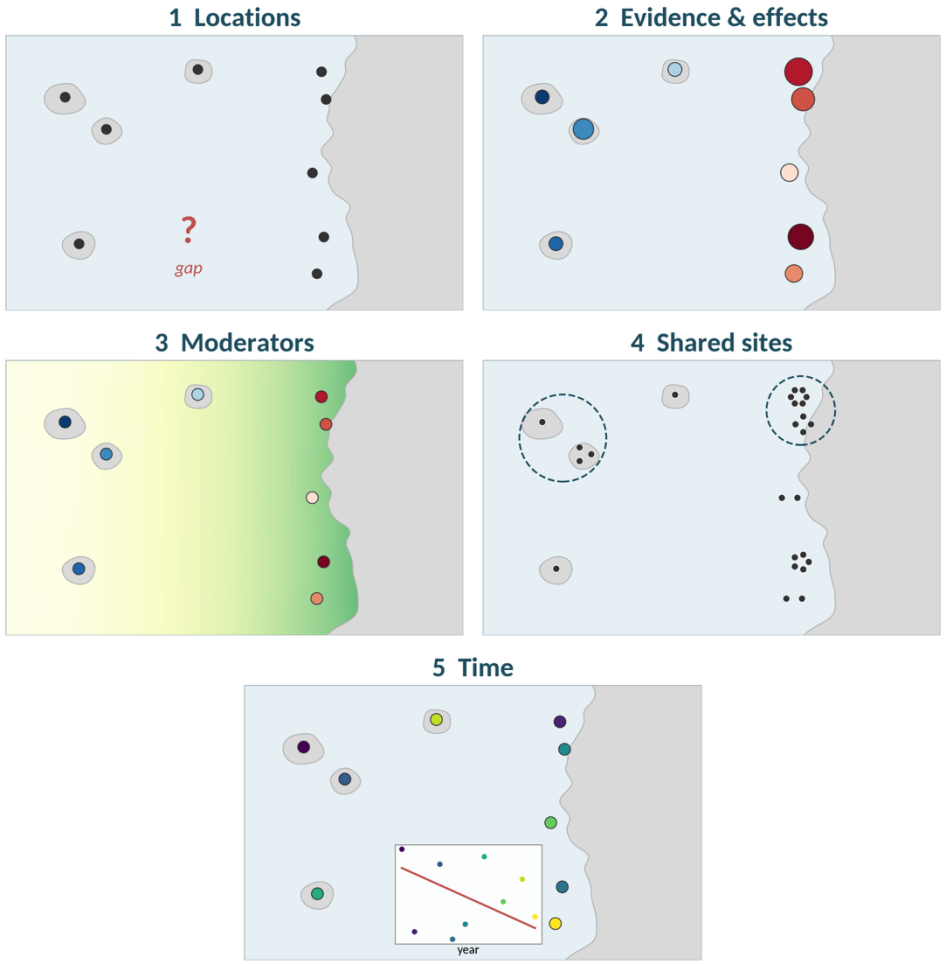




FID	Study_ID	Species	Lat	Lon
1	138812542	Passer domesticus	46.15	-0.43
2	138812556	Passer domesticus	46.15	-0.43
3	138812528	Passer domesticus	46.15	-0.40
4	138812529	Passer domesticus	46.15	-0.40
5	**138812533**	**Taeniopygia guttata**	**47.97**	**11.24**
6	138813597	Taeniopygia guttata	47.97	11.24
7	138813388	Taeniopygia guttata	47.97	11.24
8	138813415	Serinus canaria	48.90	2.21
9	138814066	Serinus canaria	48.90	2.24
10	138813195	Cyanistes caeruleus	51.27	4.48
11	138814502	Parus major	52.02	5.84
12	138812594	Taeniopygia guttata	52.16	4.49
13	138812540	Parus major	52.82	6.37
14	138813419	Cyanistes caeruleus	54.00	-2.78
15	138812560	Parus major	58.12	25.08

What can we map in a meta-analysis?

- **Study locations:** Where does the evidence come from, and what is missing?
- **Amount of evidence and effect sizes:** How much evidence, and what did each study find?
- **Moderators:** Why does the effect change between places?
- **Shared and nearby sites:** Which studies are not independent?
- **Time:** Has the effect changed over the years?



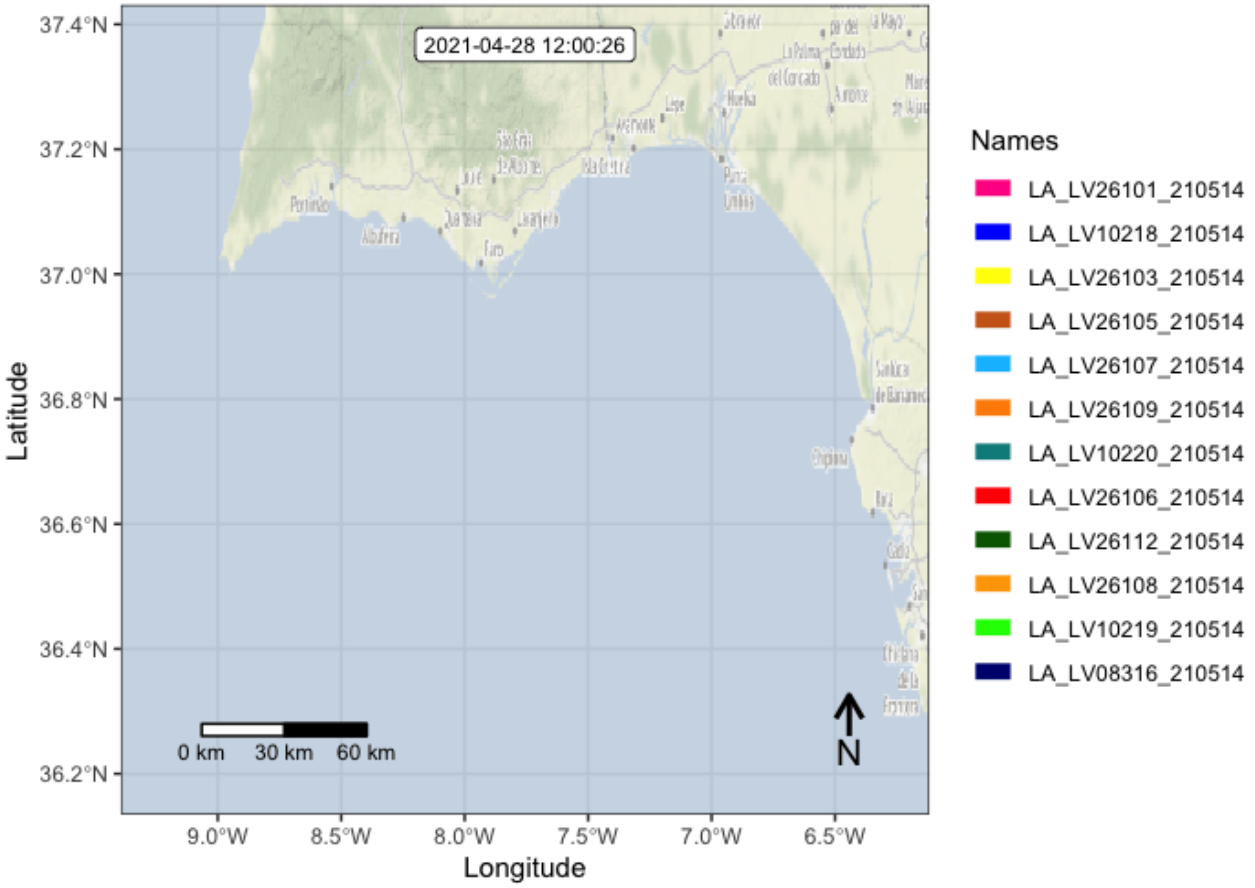
Seabirds and environmental change

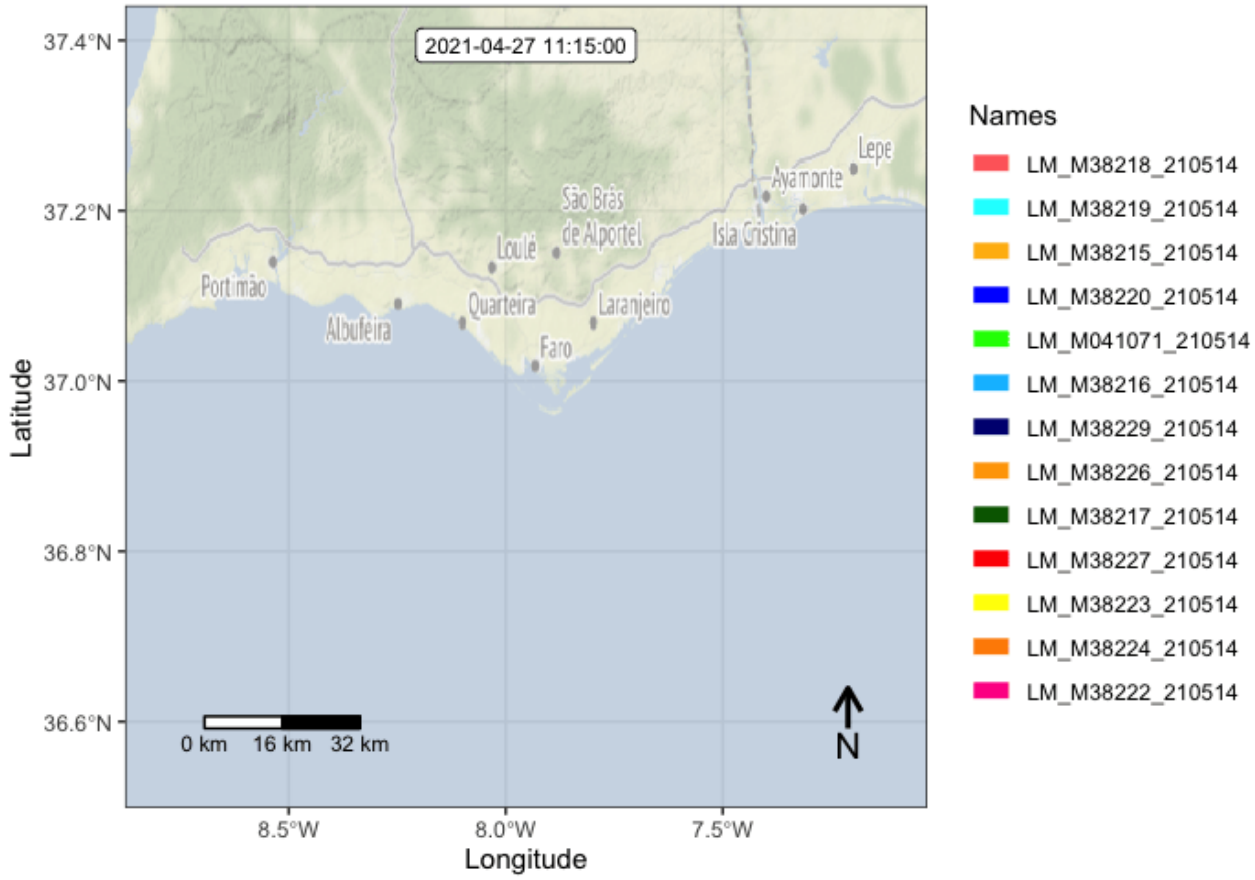




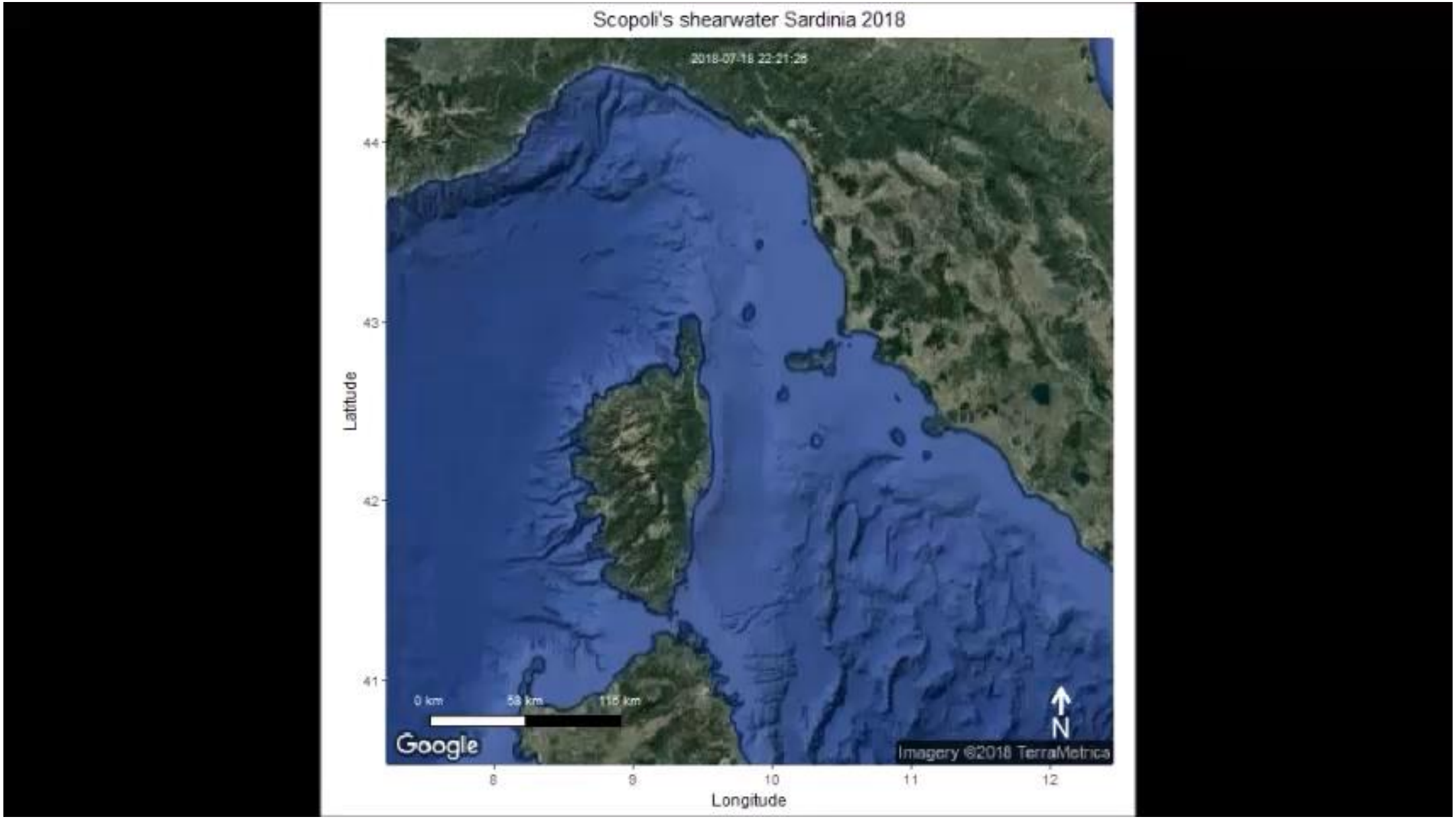


Seabirds and environmental change

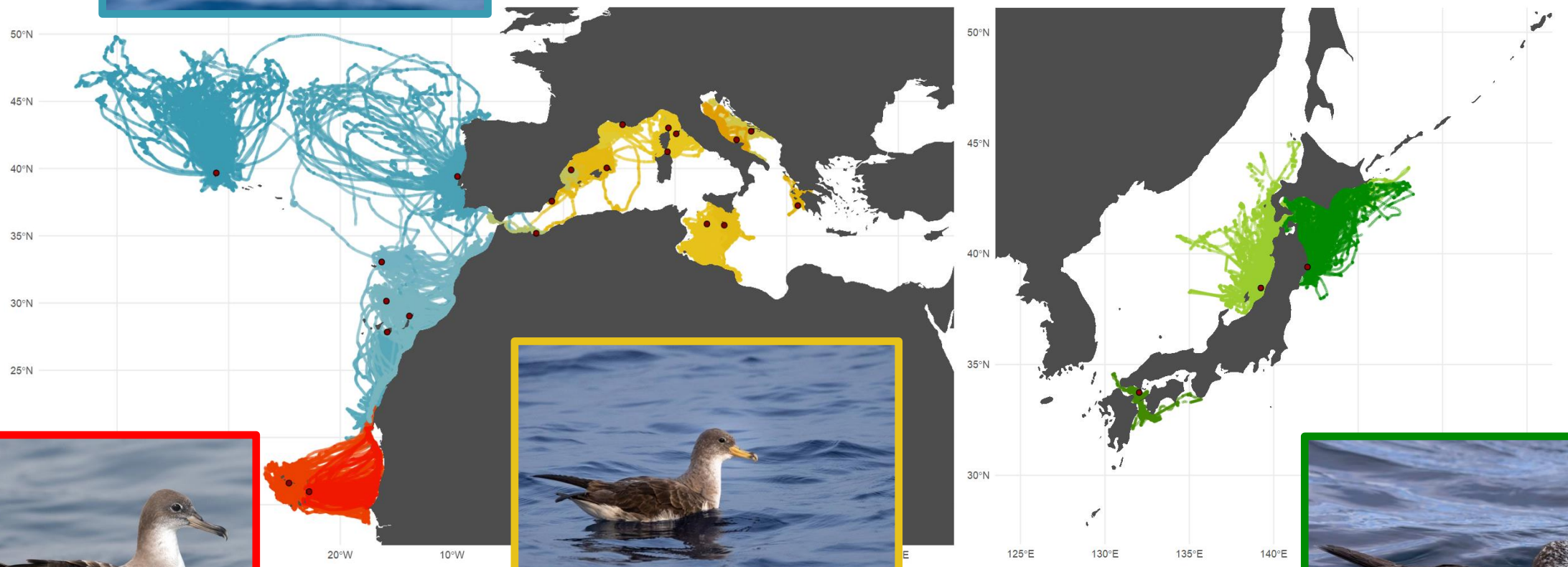




Seabirds and environmental change






50°N
45°N
40°N
35°N
30°N
25°N
20°W
10°W
50°N
45°N
40°N
35°N
30°N
125°E
130°E
135°E
140°E

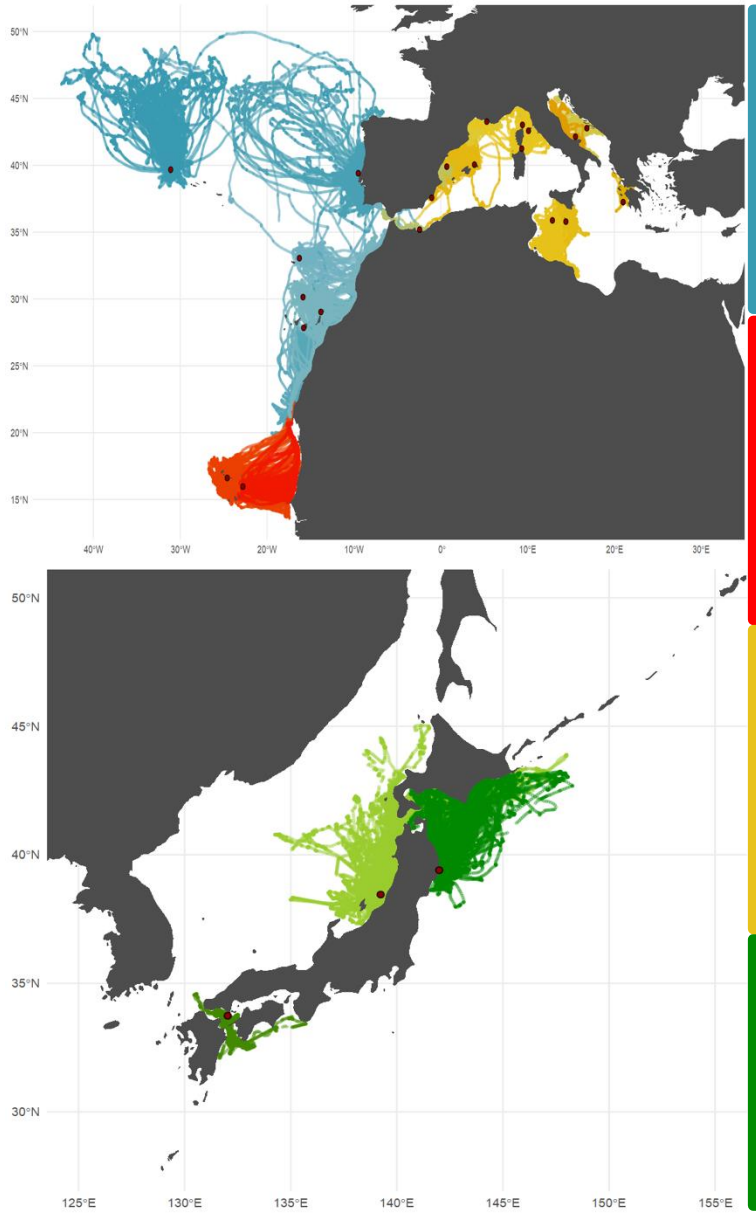






Seabirds and environmental change

- **Search string:** tracking/ movement terms and *Calonectris* species
- **Scopus:** 297 papers/ **WOS:** 329 papers
- **From all the papers:** 104 with tracking information (excluding duplicates)
- **With usable data:** 52 papers
- **Unpublished data:** 12 data owners (+25 colonies)





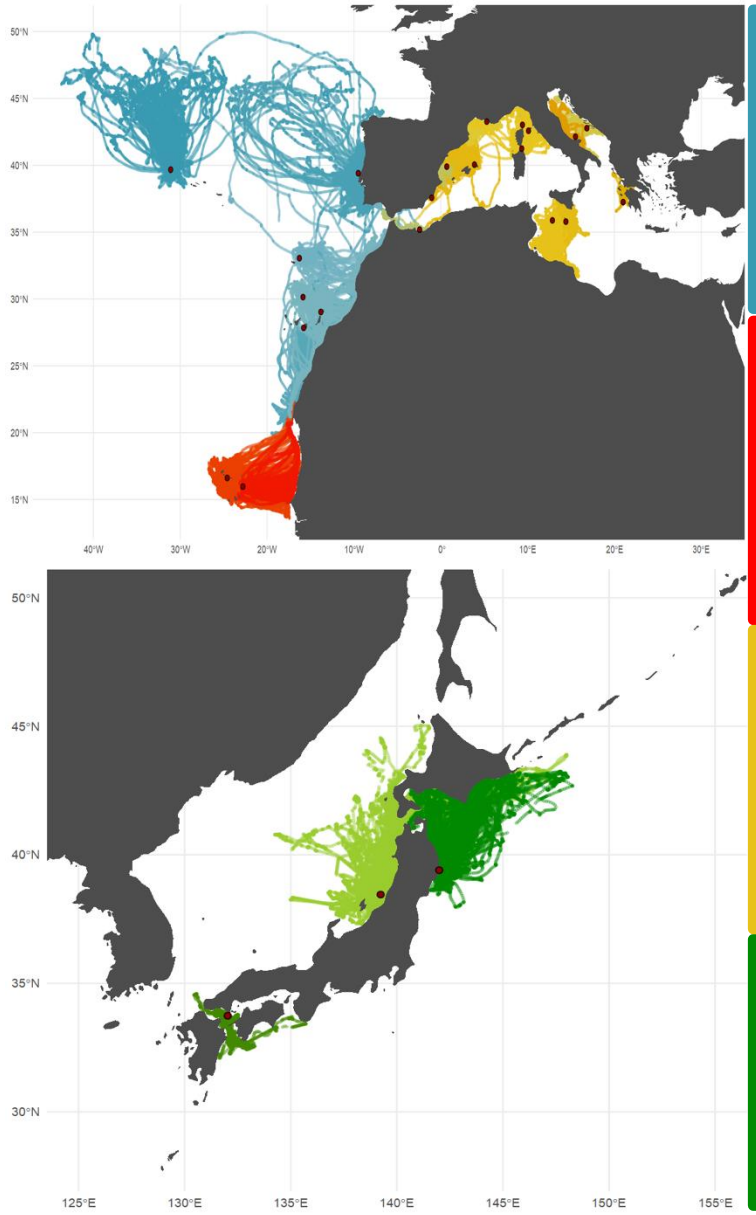






Seabirds and environmental change

- **Period:** 2000 - 2025
- **Species:** all 4 species of the *Calonectris* genus
- **Locations:** 55 colonies from 10 countries
- **Longitudinal gradient:** from the Azores in Portugal to Japan
- **Latitudinal gradient:** from Cape Verde to the Cantabrian sea
- **Coastal-oceanic gradient:** colonies located from 1 km from the coast up to ~2000km





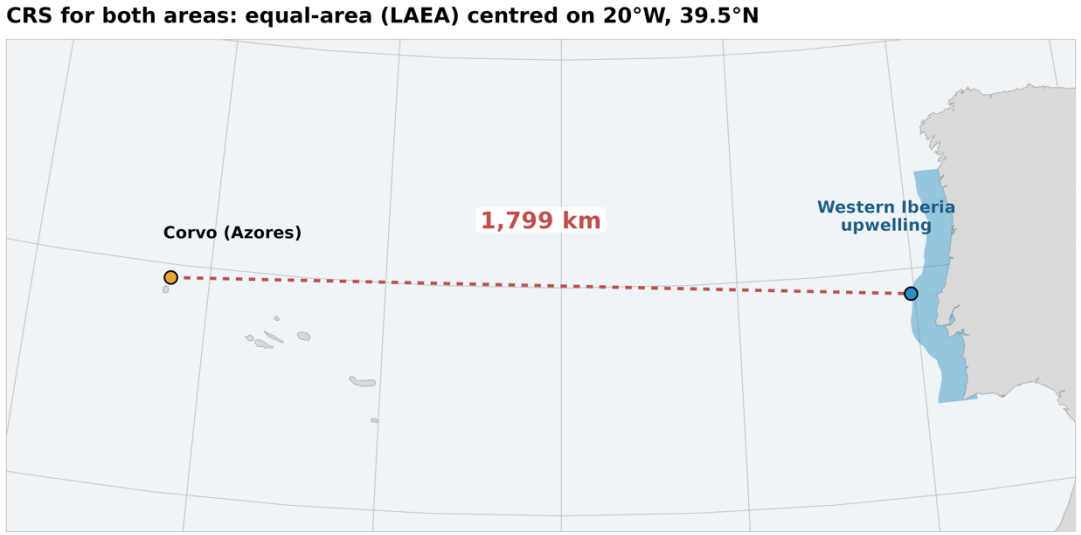






Seabirds and environmental change

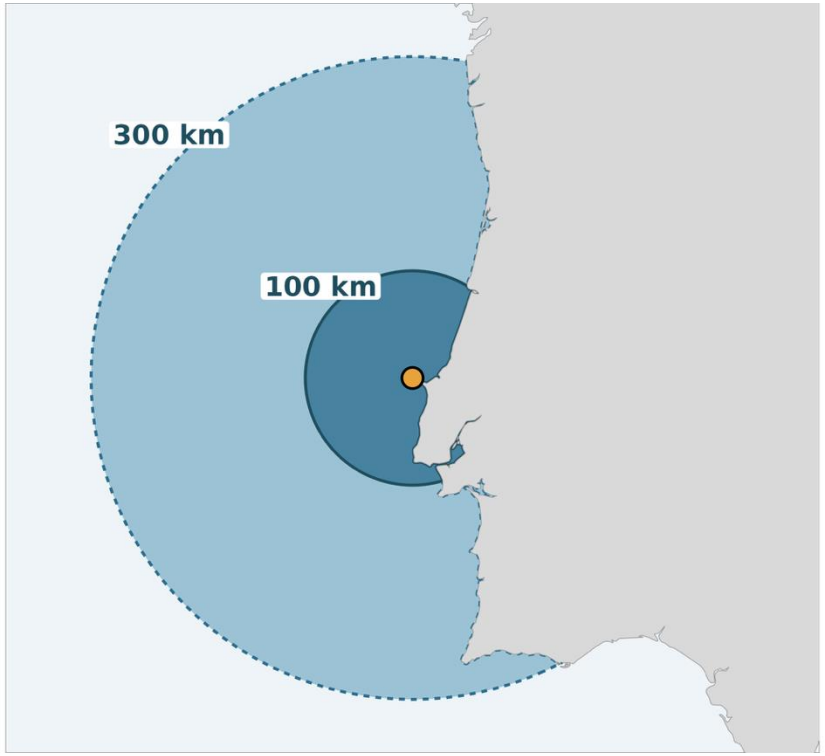
- **Distance** from each colony to the nearest upwelling system



Seabirds and environmental change

- **Distance** from each colony to the nearest upwelling system

- **Buffers** of 100 km and 300 km around each colony

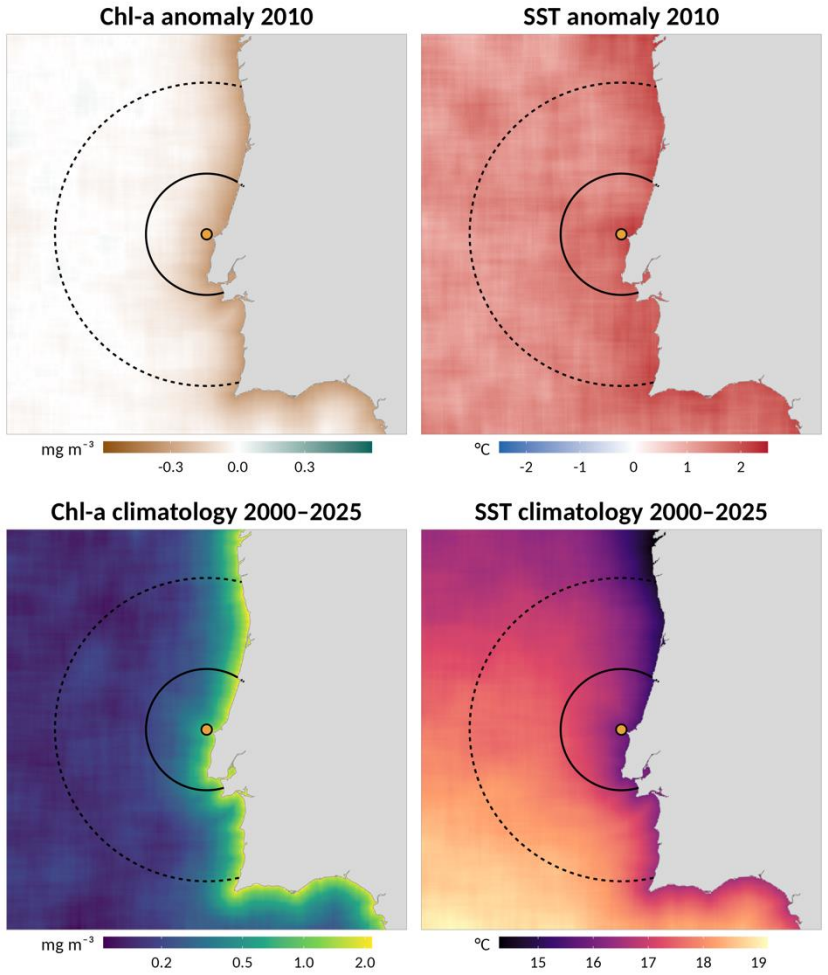


Seabirds and environmental change

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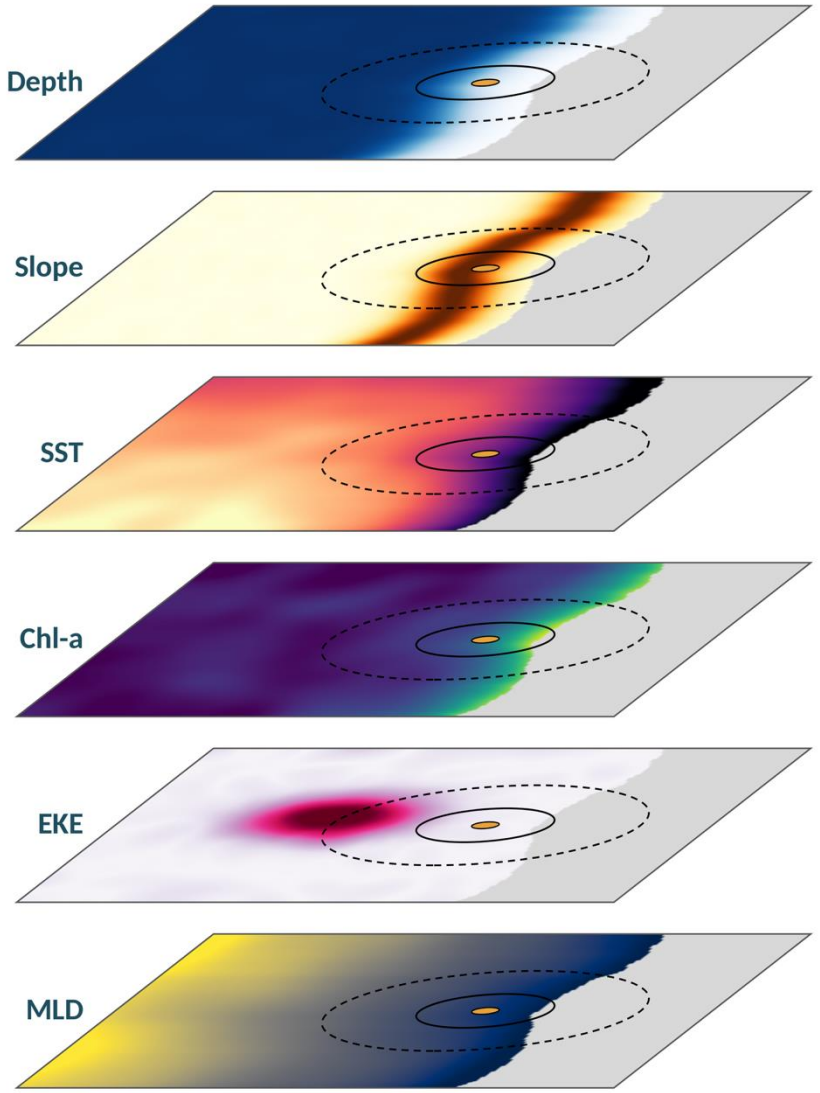
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Seabirds and environmental change

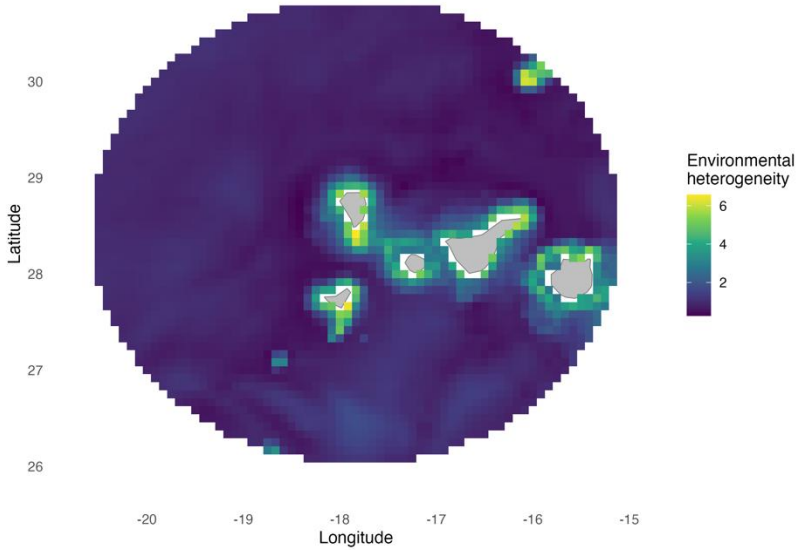
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- **Environmental heterogeneity** within each buffer, we extracted raw environmental values (not anomalies), for each colony × year × breeding phase



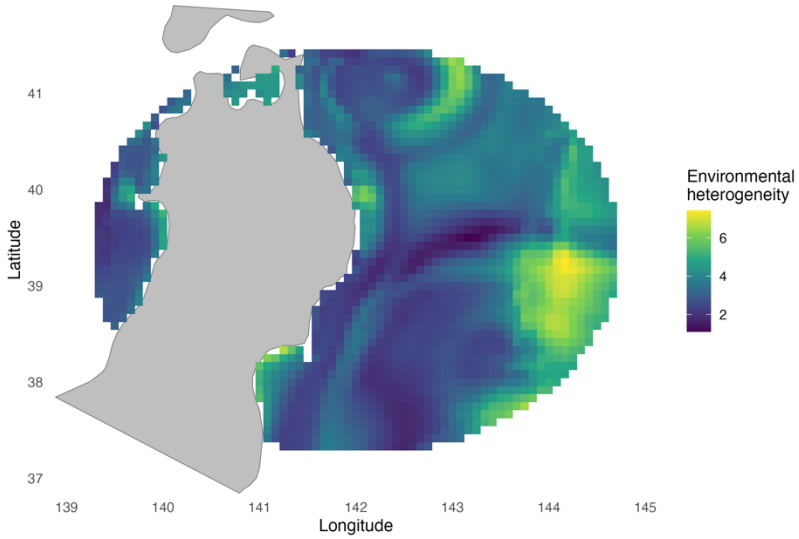
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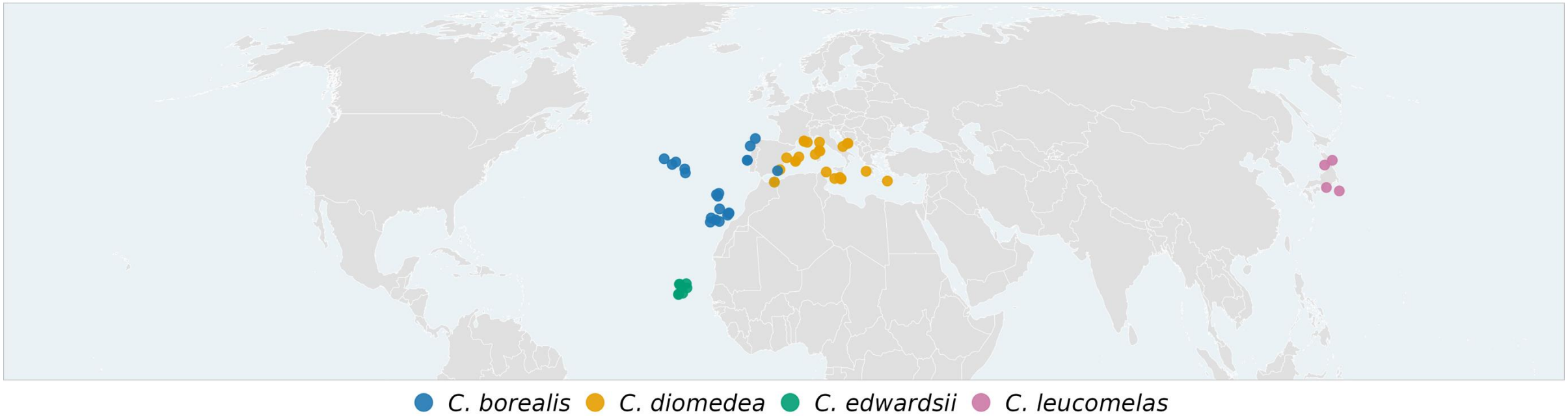
Low heterogeneity



High heterogeneity

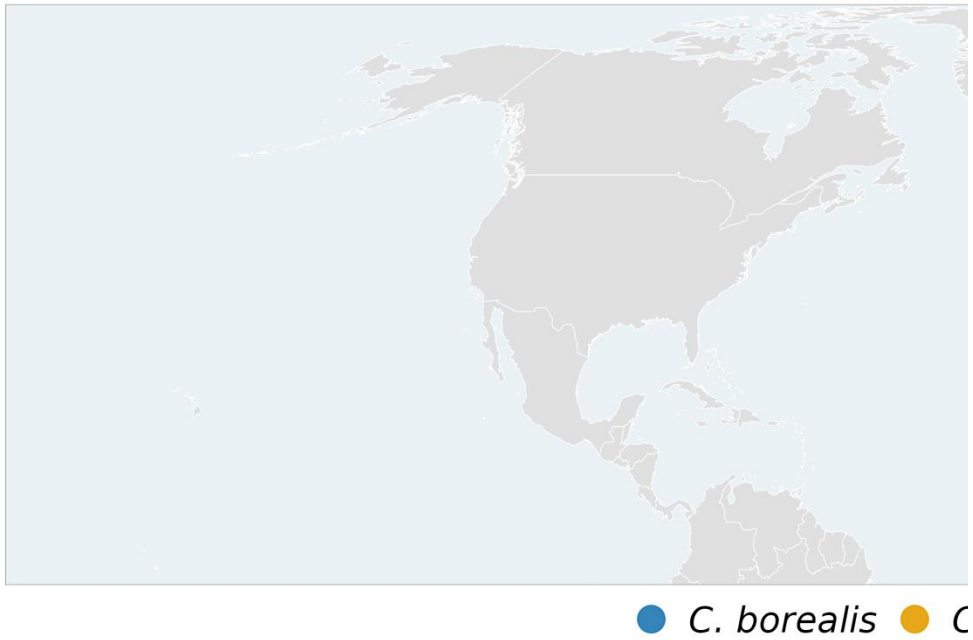


Where are the colonies?



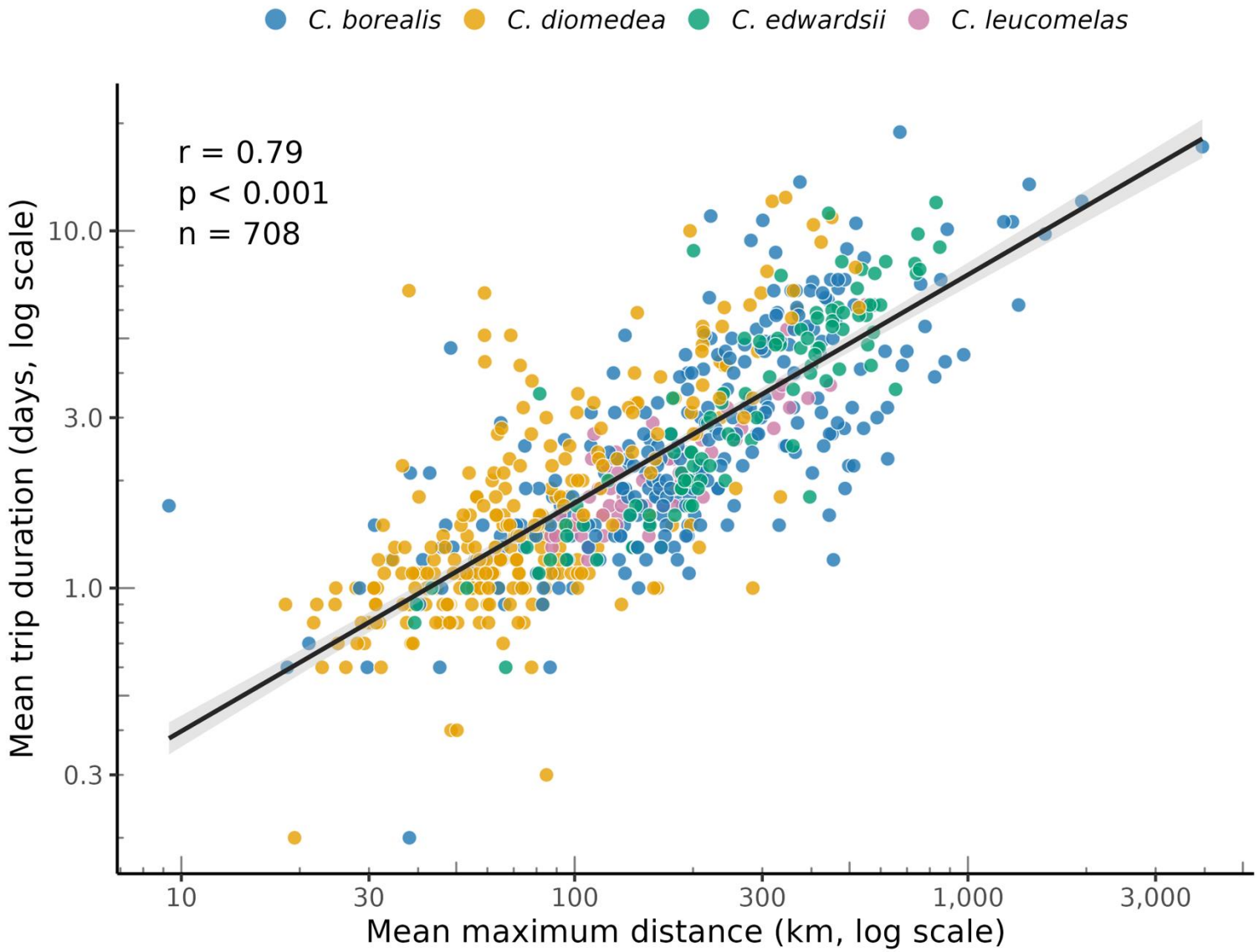
- 2000 - 2025
- *4 Calonectris* species
- 55 colonies (52 published papers and 12 data owners)
- Global distribution of the genus

Checking the raw data

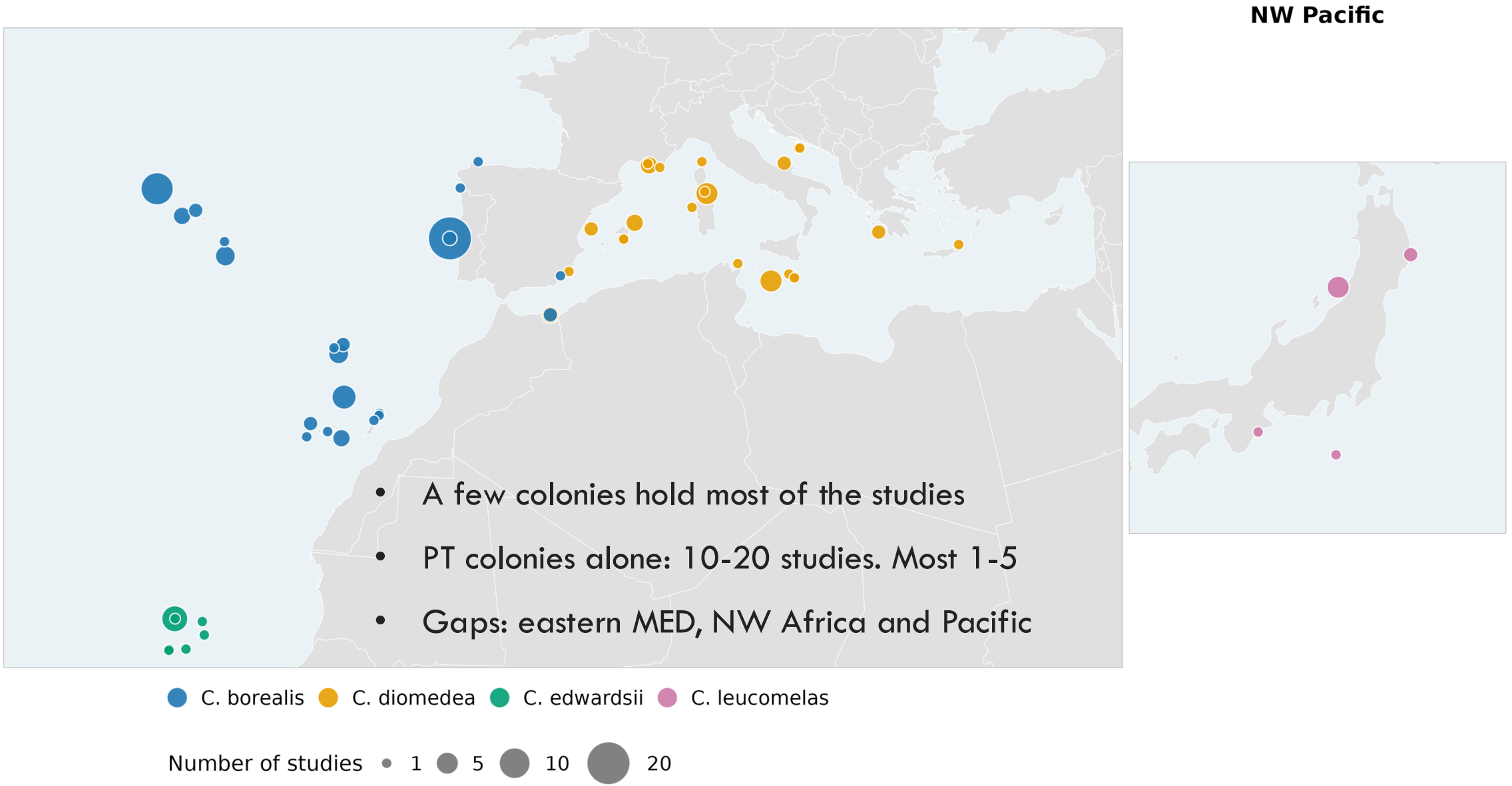


Longer trips > Longer distances

Foraging effort

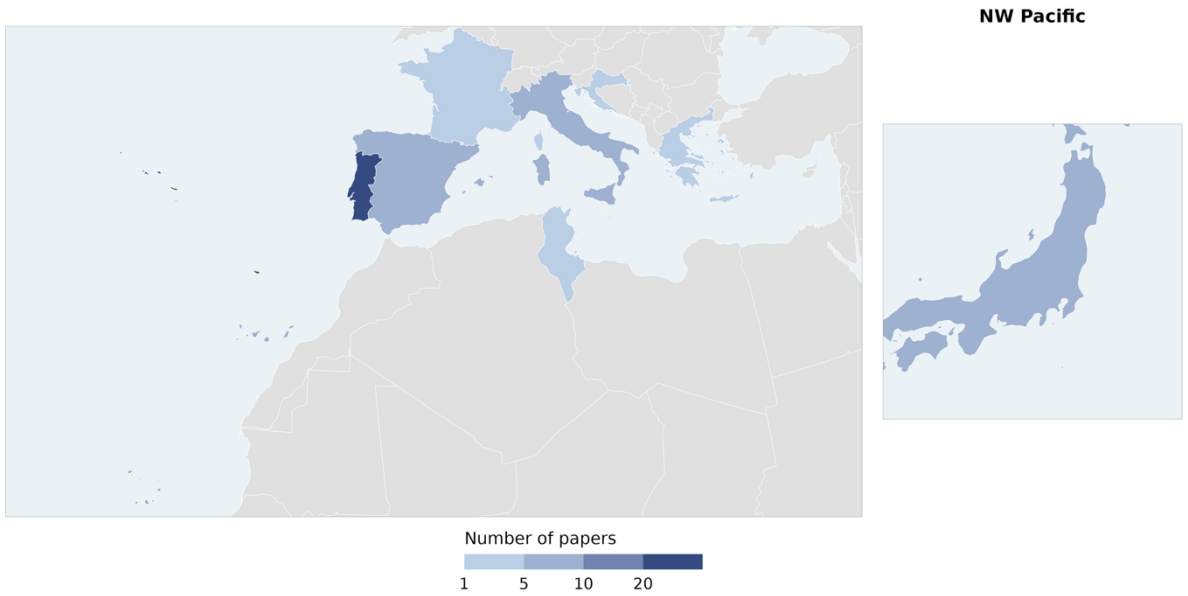


Where does the evidence come from?



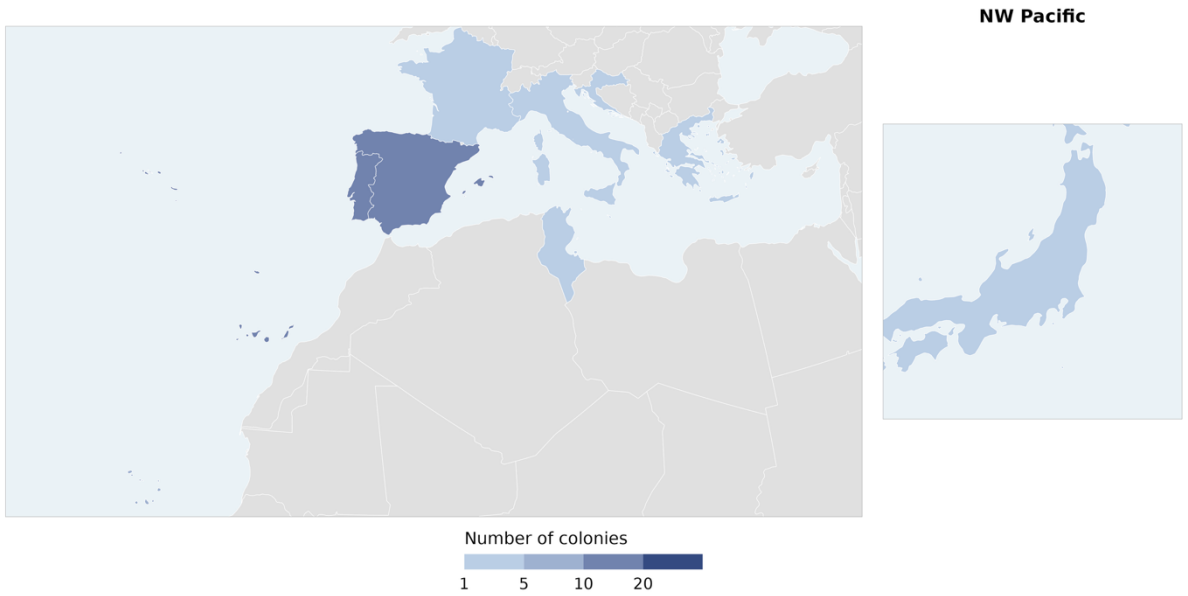
Who does the research?

Papers per country



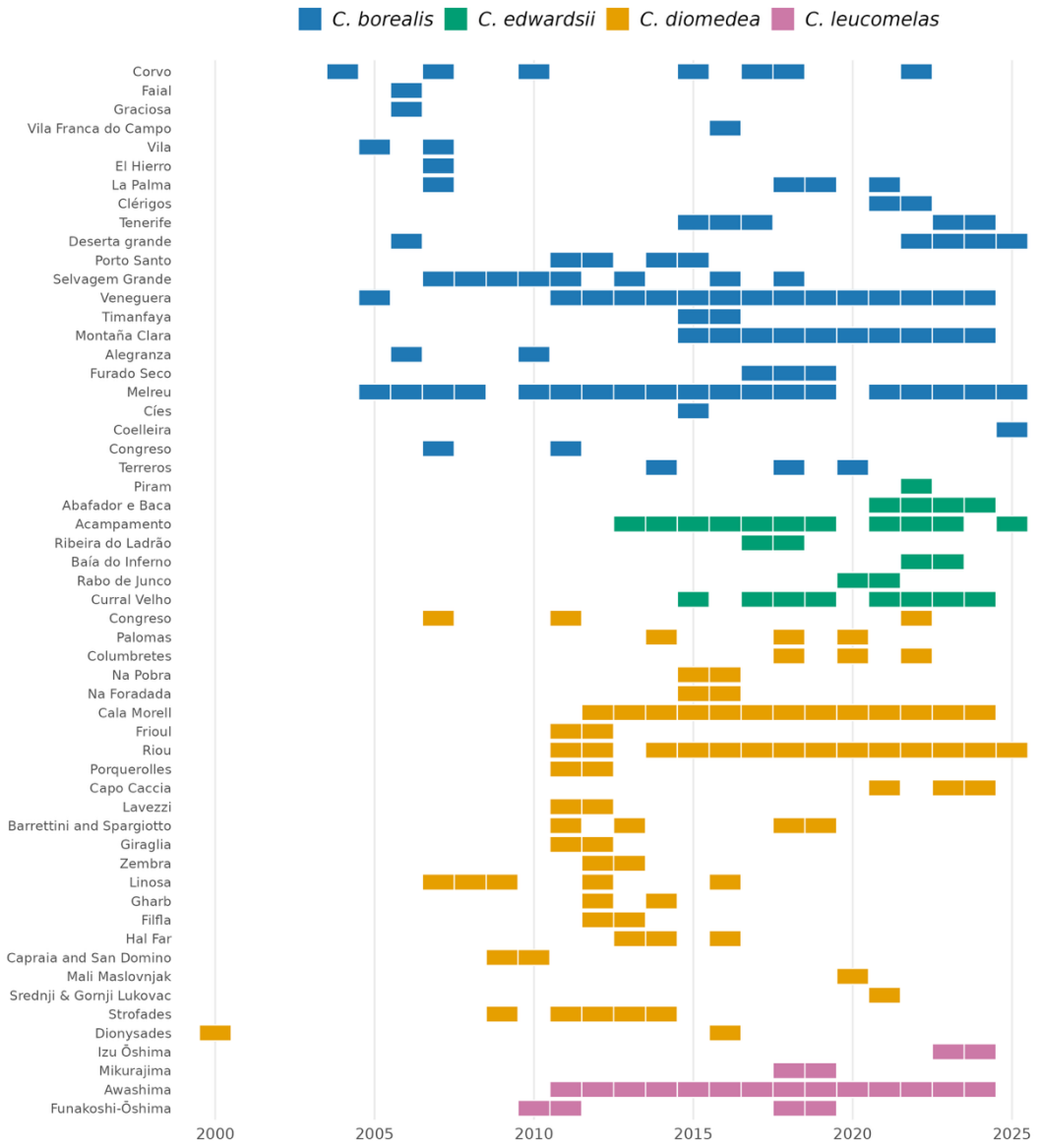
- Portugal leads: 27 papers
- Then Spain (10), Cape Verde (9), Italy (8), Japan (7)
- Most papers come from one colony: Berlenga (20)

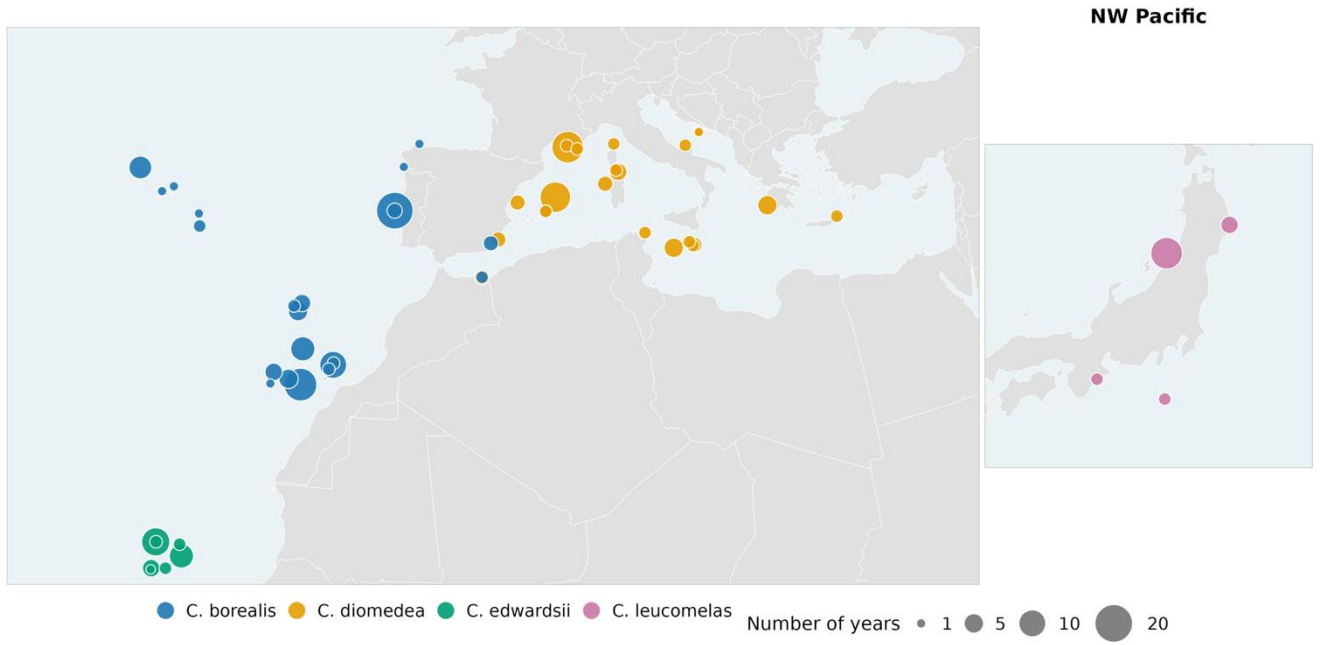
Colonies per country



- Spain has the most colonies (16) but only 10 papers
- Portugal has 11 colonies and 27 papers
- More colonies does not mean more studies

When were the colonies tracked?

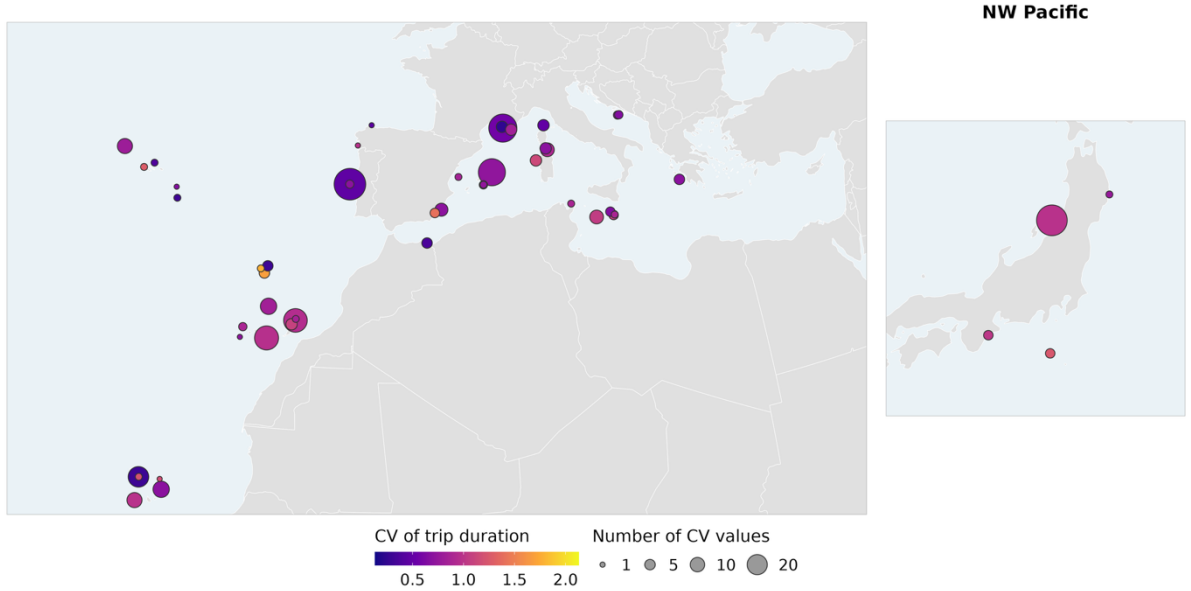




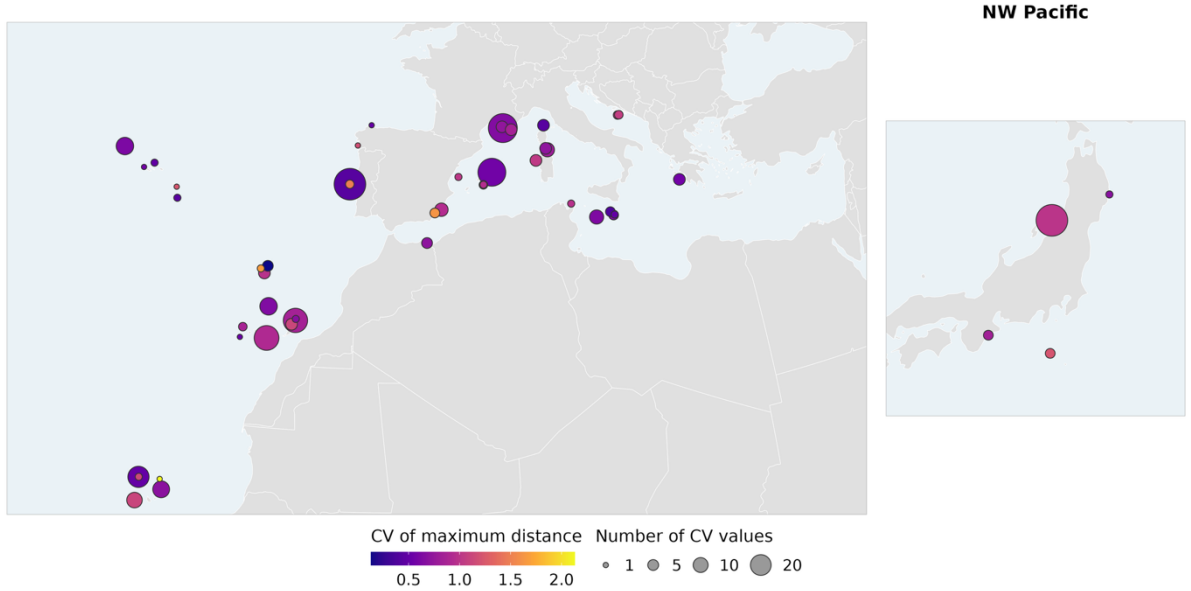
- Only 7 of 55 colonies have ≥ 10 years
- 9 colonies: a single year only
- Different colonies tracked in different periods

How much evidence and where?

Trip duration



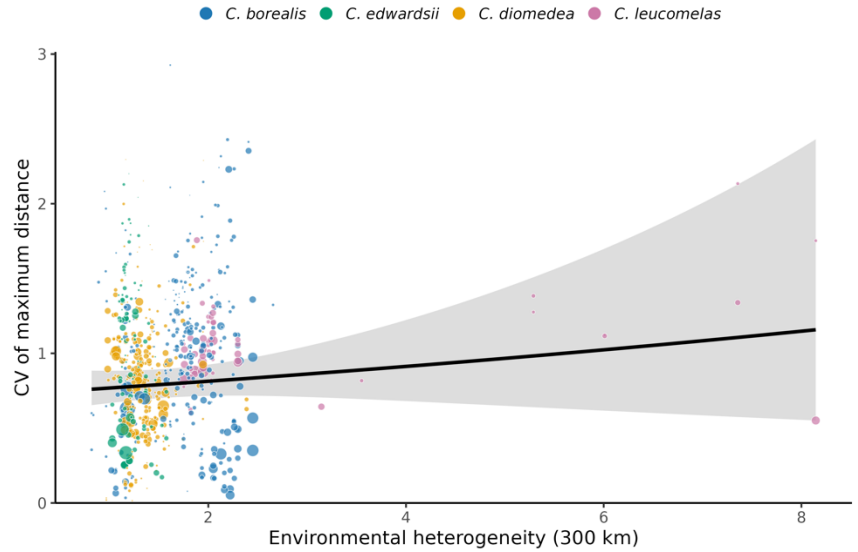
Maximum distance



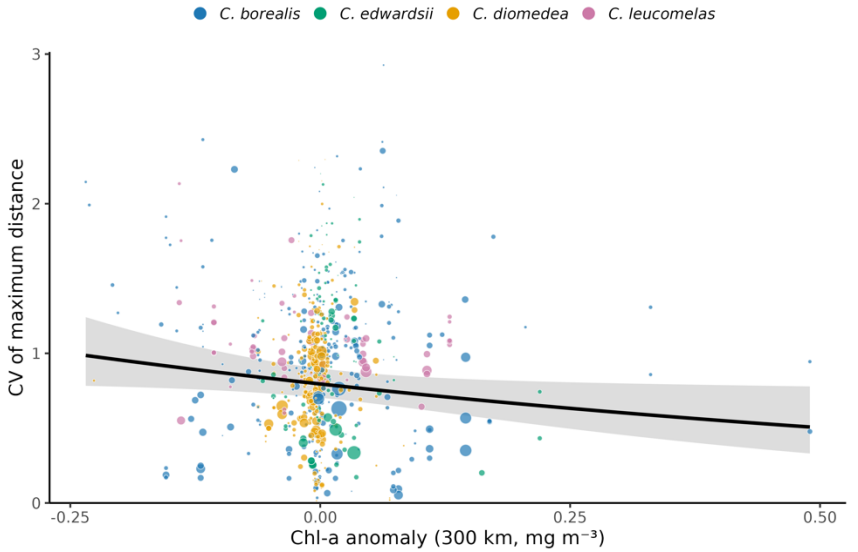
- **Coefficient of Variation:** SD / mean
- **Effect size:** log CV, adjusted for number of trips
- **Low to medium variability (CV < 0.85):** Berlengas and Mediterranean colonies
- **High variability (CV ≥ 0.85):** Canary Islands and Japanese colonies

Why does the effect change across space?

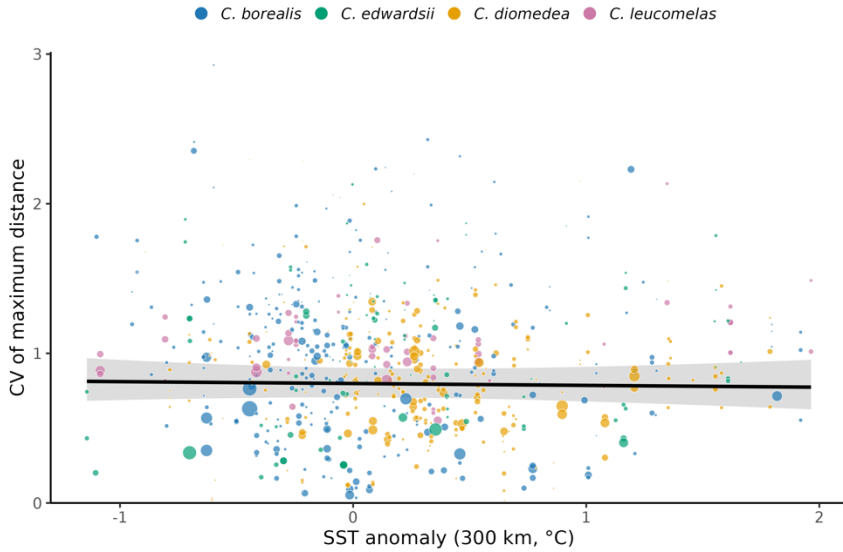
Env. heterogeneity

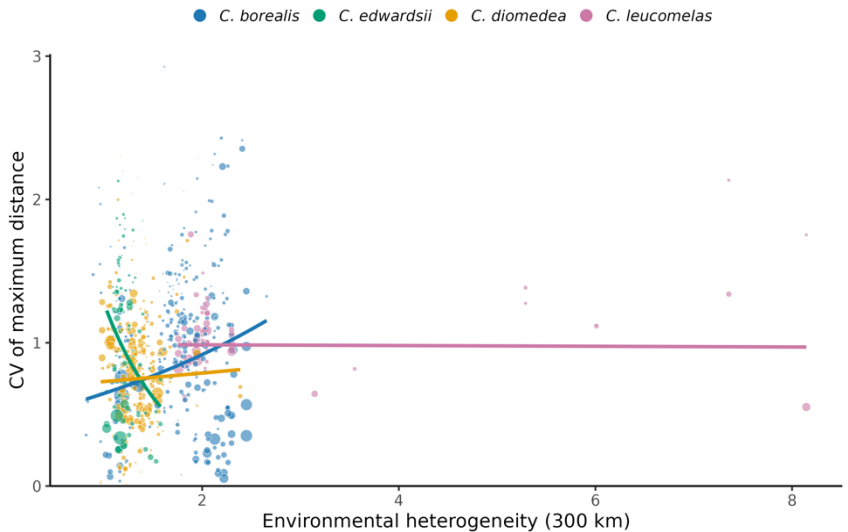


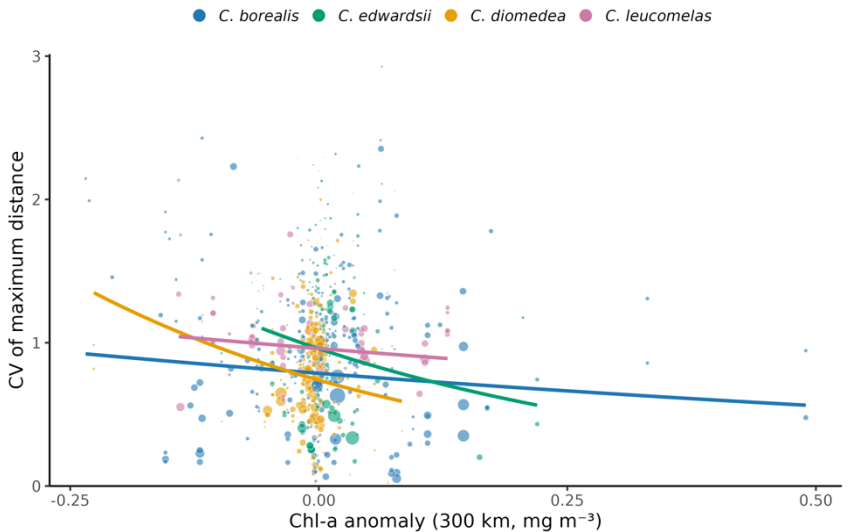
Chl-a anomaly

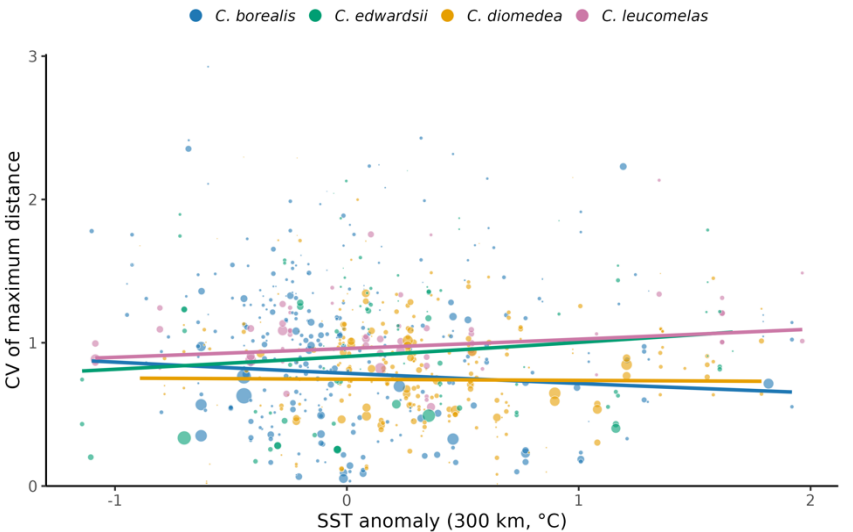


SST anomaly

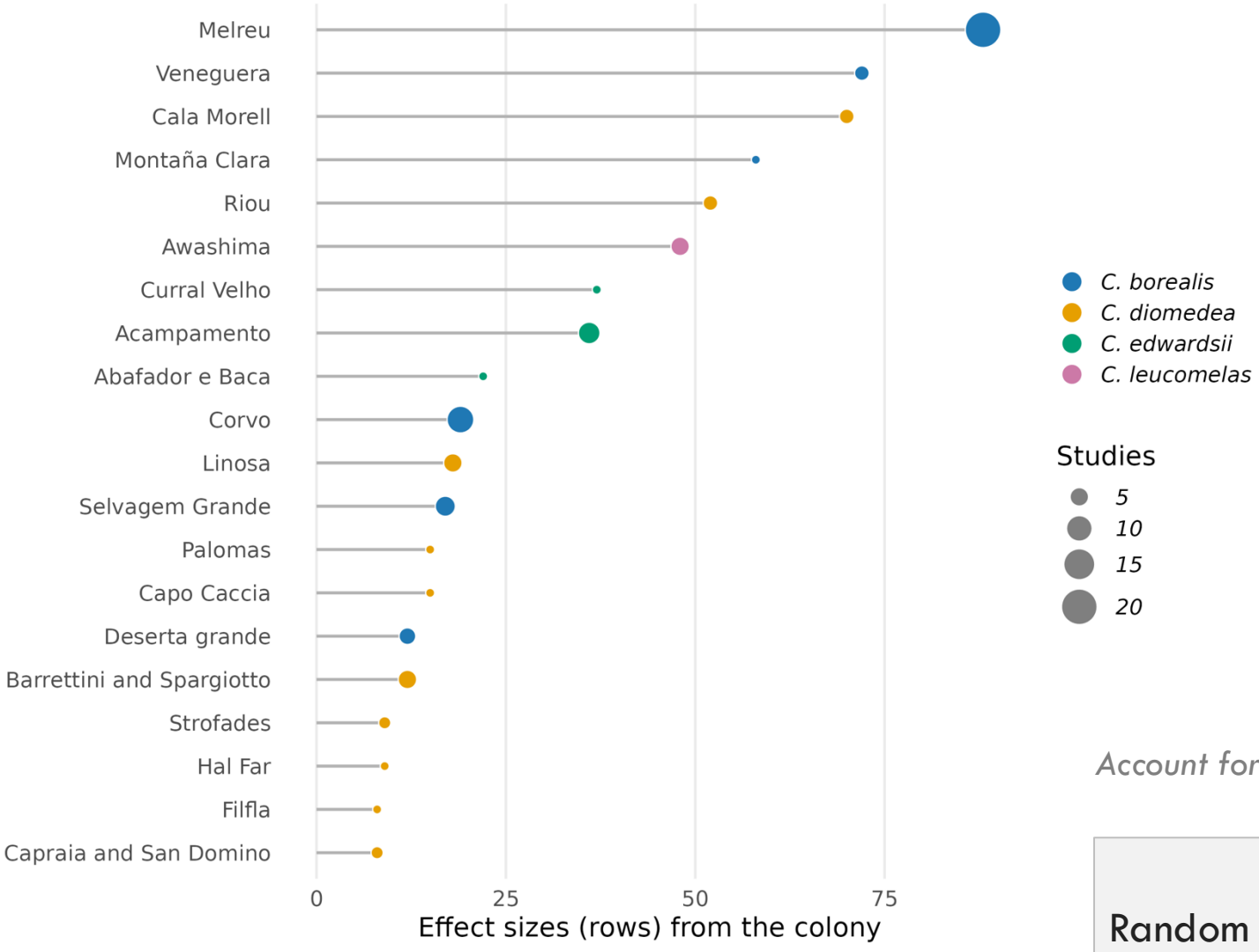








Which studies are not independent?



- Berlenga: 88 values from 20 studies
- Canary Islands: 58 values from 1 study
- Same colony, same birds, different years: no independence

Account for it: year within colony, and study as random effects

Random term = (1 | Colony / Year) + (1 | Study_ID)


Sea of NAs
Isles of
Missing Coordinates
Start:
data.csv
Your first map
X
Cape of
the Wrong CRS
Strait of
Non-Independence
N
Here be
swapped lat & long
Thank you!
Jorge Pereira
(jorge.pereira@uc.pt)
Isles of One Study Only
CRS: unprojected WGS 84. Pirates never cared about distances.